Unsupervised Wavetable Seam Artifact Repair via Hybrid GA–PPO Meta-Search

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Julian M. Kleber
Independent Researcher
Email: julian.m.kleber@gmail.com

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Code (ReelSynth engine + DenoiseOpt meta-search): https://github.com/reeldemo/reelsynth, https://github.com/reeldemo/denoise-opt-meta

ABSTRACT

In single-period wavetable synthesis, a sample discontinuity at the wrap boundary injects a broadband transient that repeats at the fundamental, producing an audible periodic click. We treat this *cycle-local seam artifact* with **DenoiseOpt**: a residual-scored hybrid meta-search that selects and fits bake-cell operators Θ at the period boundary. The bake cell can compose DualCosine crossfades, FIR blends, and optional lightweight residual networks, with optional Mixture-of-Experts (MoE) soft gating. Paired studio-clean recordings are not required. Candidates maximize the *prolonged residual score* $R \in [0, 1]$ against a procedural ideal sibling $r^\star = G(\text{seed}, \text{cliff=off})$. The cracked engine is $x = G(\text{seed}, \text{cliff=on})$ from the same draw. No-bake leaves x unchanged and scores that unrepaired period against $r^\star$. No-bake is not the ideal. DualCosine may center PPO advantages only. Ranking still maximizes absolute R. The outer loop interleaves genetic algorithm (GA) tournament selection, PPO architecture proposals, and optional population-based training (PBT). On a frozen canonical sine+cliff holdout (seed `20260719`), the compact meta-favorite reaches $R \approx 0.991$ versus classical no-bake $R \approx 0.972$, `seam_fir3` $R \approx 0.961$, and DualCosine $R \approx 0.825$. Hard-cliff strata (top 10% wrap-jump, n=410) keep favorite $R \approx 0.9915$ while DualCosine falls to $R \approx 0.657$. No-bake R stays high (≈ 0.967) because residual mass is small relative to ideal RMS (passthrough $x \mapsto x$). A same-generator corrupt→corrupt Noise2Noise control reaches $R \approx 0.992$. A supervised LSTM sibling ceiling reaches $R \approx 0.995$. Wavetable-native transfer uses true ReelSynth-exported factory periods (n=25) and external AKWF CC0 single-cycle WAVs (n=24) under the same wrap protocol (no LibriSpeech/MUSDB). On exports the favorite trails no-bake / FIR (domain shift). On AKWF it matches no-bake and beats DualCosine. A classical-board sci/eng wrap transfer (CWRU, MFPT, MIT-BIH, PTB-XL subset, synthetic CNC/PMU) is reported in the main Results with honest not-executed deep-SOTA rows. Scope is cycle-local wavetable seam restoration (DAFx / AES / arXiv cs.SD). We do not claim speech enhancement or deep SOTA on sci/eng transfer boards.

Keywords unsupervised learning; wavetable; wrap discontinuity; seam restoration; DenoiseOpt; prolonged residual score; neural architecture search; reinforcement learning; genetic algorithms; meta-learning

1 Introduction

Wavetable oscillators store a single period of a waveform and advance a read pointer modulo the table length. When $x[0] \neq x[L-1]$, the wrap is a hard discontinuity that, under cyclic playback, becomes a periodic click with a broadband spectrum locked to the fundamental. The same geometry appears whenever a short loop buffer is tiled for listening or for objective evaluation. We define the repair task as *cycle-local seam restoration*: edit a neighborhood of the wrap (and optionally a lightweight residual cell over the period) so that multi-period playback matches an ideal reference that never opens the wrap cliff. That outer agreement is scored by the prolonged residual R for this periodic seam artifact class.

Figure 1 shows one real sine-wrap edge case from the frozen canonical holdout (same `make_batch` tile, seed `20260719`). Four roles must stay distinct: (a) ideal sibling $r^\star$ (cliff withheld; unscored target); (b) no-bake / cracked engine x after open-wrap cliff ($R \approx 0.950$ vs $r^\star$; ideal overlay); (c) DualCosine classical bake on that same tile ($R \approx 0.603$ on this severe cliff); (d) Ours / DenoiseOpt favorite bake (`champion_iter_000235`, tile $R \approx 0.9917$). Panels (e)–(f) plot the modulation residual and a seam zoom so Ours is readable against DualCosine. No-bake is not the ideal. DualCosine is one classical row and, separately, a PPO advantage baseline (Section 3).

Classical virtual-analog practice already supplies seam-local tools. Alias-free BLIT/BLEP-style synthesis [1], PolyBLEP / fractional-delay antialiasing [3], and BLAMP corner corrections [2] reduce wrap and geometric discontinuity energy in-band. Those operators act locally. We score cycle-local bake approximations of them beside DualCosine (Section 5.7). Open questions remain: which bake parame-

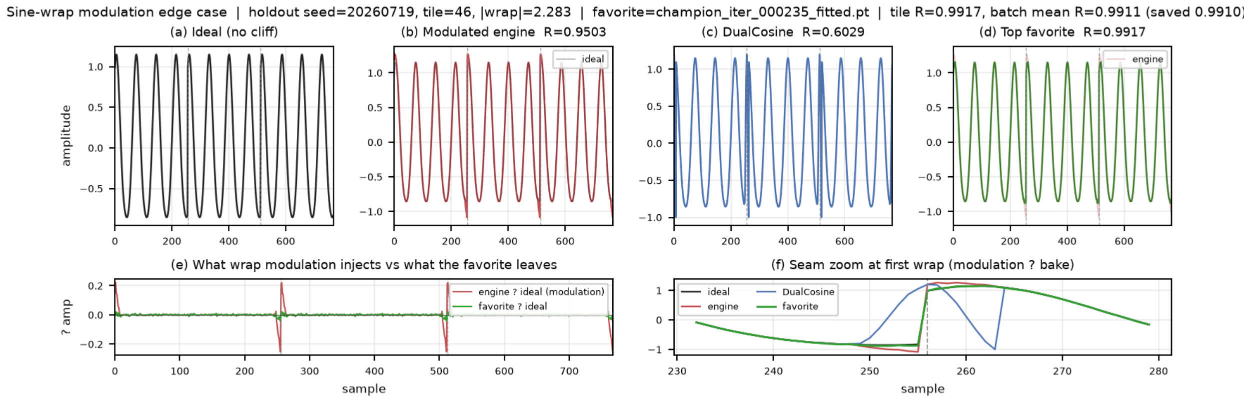

Figure 1: Canonical holdout sine-wrap edge case (seed `20260719`, L=256, N=16 tiles, same tile across panels). Distinguish four roles: (a) ideal sibling $r^\star$ (cliff withheld; scoring target, not a method); (b) no-bake / cracked engine x (passthrough Θ=id, scored vs $r^\star$); (c) DualCosine classical end-fade bake; (d) Ours (DenoiseOpt favorite bake). (e)–(f) Modulation residual and seam zoom. Prolonged residual $R \in [0,1]$ (1 = best) is printed per panel where scored. Accessibility: six-panel waveform comparison of ideal, no-bake, DualCosine, and Ours.

ters, and which residual cell if any, minimize prolonged tiled error against a no-cliff ideal. Label-free restoration work such as Noise2Noise shows that useful reconstructors can be ranked without paired clean targets when the evaluation geometry is explicit [4]. Speech-domain Noise2Noise denoisers extend that idea to audio waveforms [5]. Mixture-invariant unsupervised separation likewise avoids paired clean stems for ranking [29]. On the search side, NAS surveys [16], RL controllers [17, 18], aging evolution [20, 21], population-based hyperparameter schedules [22], CMA-ES tutorials [24], and evolution-guided policy gradients [23] motivate a hybrid outer loop: GA for population diversity, PPO for discrete architecture edits [19], and PBT-style exploit–mutate on elites. Platform-aware and progressive NAS add compute-aware search priors [27, 28]. MoE soft gates suggest routing among heterogeneous tiny cells under one residual score [25]. Waveform separation nets (Wave-U-Net, Conv-TasNet, dual-path RNN, SepFormer) and audio DL surveys inform the searchable cell family at bake-scale width [7–9, 11, 12, 30, 31], with U-Net / ResNet / attention primitives as block ancestors [10, 13–15].

This paper contributes a meta-learning protocol for this periodic seam instance. An outer RL+GA(+PBT) loop proposes bake-cell operator graphs and continuous bake parameters. Each candidate is scored by prolonged residual $R \in [0,1]$ (1 = best) between engine-baked multi-period audio and a procedural ideal sibling $r^\star = G(\text{seed}, \text{cliff=off})$. Paired studio-clean wavetables are not required for every trial. Differentiable DSP modules [6] place DualCosine / FIR / MLP seam ops in an interpretable stack. Bayesian HPO [26] informs prior families that compete under the same R. A CUDA hybrid search campaign (PPO+GA+PBT+NAS, depth and MoE biases) has passed a 5,000-iteration clean gate with champion $R \approx 0.9909$ versus DualCosine ≈ 0.8166 on the runner geometry. The frozen canonical holdout and inference benches in Section 5 are the measured evaluation matrices for this report.

Primary claim scope is cycle-local wavetable seam restoration (DAFx / AES / arXiv cs.SD). We do not claim speech enhancement, music source-separation, or deep SOTA on classical sci/eng transfer boards.

1.1 Contributions

1. **Problem framing.** Cast wavetable wrap discontinuity as cycle-local seam restoration under prolonged tiled evaluation, separating seam-local diagnostics from residual R.

2. **Hybrid DenoiseOpt outer loop.** Specify RL+GA(+PBT) search over discrete operator graphs and continuous bake parameters, with named branches (GA crossover, PPO mutation, PBT exploit, MoE soft gates).

3. **Formal residual protocol.** Define bake operator Θ, ideal-sibling generator G, outer objective $\max R(y, r^\star)$, and reimplementable Algorithms 1–9 (Appendix A). No wrap-closure or search-convergence guarantees.

4. **Open evaluation path.** Release companion code for the synthesizer engine and meta search, with frozen 5k-gate figures, hard-cliff strata, N2N/seq baselines, and wavetable-native transfer benches.

Section 2 organizes related work by mechanism. Sections 3–4 detail the residual objective and search protocol. Section 5 reports the 5k-gate freeze and inference benches.

2 Related Work

We organize prior work by mechanism. **Used** lines are methodological or comparative anchors for residual-scored RL+GA meta-search. **Screened** lines were inspected as contrast and are excluded from the dependency set. Paraphrases prefer attached abstracts and PDF snippets over title-only citation.

2.1 Discontinuity control and modular audio DSP

Alias-free digital synthesis of classic analog waveforms (BLIT/BLEP lineage) states the wrap discontinuity problem for trivial oscillators [1]. PolyBLEP-style fractional-delay corrections give practical antialiasing oscillators for subtractive synthesis [3]. BLAMP corner rounding treats discontinuities in the first derivative (triangular edges, clipping, rectification) without heavy oversampling [2]. We score and illustrate cycle-local approximations of those three operators beside DualCosine on the same prolonged-R protocol (Figure 10, Section 5.7). They are no longer citation-only anchors. DDSP reframes DSP modules as composable blocks inside learning pipelines [6]. We use that framing for small bake/FIR/MLP seam cells. We do not train end-to-end neural vocoders for every meta trial. These works are **used** as domain anchors for the operator vocabulary, as scored classical controls, and for why prolonged cyclic playback is the outer judge.

2.2 Label-free restoration

Noise2Noise shows that restoration mappings can be learned from corrupted observations alone by exploiting structure in the corruption process [4]. Speech Noise2Noise denoisers apply the same idea to waveforms without clean paired audio [5]. Mixture invariant training ranks unsupervised separation systems without stem labels [29]. That philosophy motivates ranking denoise candidates without paired studio-clean wavetable cycles. We train a true corrupt→corrupt N2N-style seam restorer on the same L=256 `make_batch` geometry (primary baseline), plus a sibling-supervised ceiling and lightweight LSTM / 1D-CNN seq models (Section 5). Prior speech N2N restorers target broadband noisy speech. Here the domain is periodic seam cliffs with procedural siblings. We **use** this family as conceptual justification for unsupervised residual ranking and as an explicit learned contrast to DenoiseOpt meta-search. Full Demucs / DiffWave stacks are screened: domain mismatch (full-band speech/music denoisers vs short periodic seams), compute, and OA/eval protocol mismatch. We do not retrain image Noise2Noise CNNs.

2.3 Waveform denoising and separation cells

Deep learning for audio surveys the move from hand-designed features to learned time/frequency models [7]. Wave-U-Net performs end-to-end source separation with multi-scale 1-D U-Nets [8]. Improved Wave-U-Net speech enhancement and singing-voice U-Nets popularized encoder–decoder skips on audio [9, 10]. The original U-Net, ResNet residuals, and Transformer attention supply the block primitives we shrink to bake width [13–15]. Conv-TasNet and dual-path RNN demonstrated strong time-domain separation with temporal convolutions and long-context paths [11, 12]. SepFormer and audio spectrogram transformers push attention-heavy audio stacks further [30, 31]. These lines are **used** as design priors for tiny searchable cells (shallow U-Net / dilated / dual-path / attention blocks under a strict per-trial bake budget). They are not frozen full-scale separators inside every meta trial.

2.4 NAS and RL controllers

Elsken et al. taxonomize NAS by search space, strategy, and evaluation cost [16]. Their performance-estimation axis includes lower-fidelity proxies such as shorter training, data subsets, and lower-resolution images [16]. Zoph and Le couple RL controllers to architecture search [17]. ENAS adds parameter sharing for efficient discrete search [18]. Progressive and platform-aware NAS add staged evaluation and latency-aware rewards [27, 28]. We **use** this family as the ancestor of RL proposing architecture edits (add/remove/mutate ops, toggle residual-primary, mutate MLP/FIR), scored with prolonged residual. PPO supplies the clipped-surrogate policy update when the hybrid search loop logs PPO mutation branches [19].

2.5 Evolutionary NAS and population schedules

Large-scale and regularized / aging evolution showed that evolutionary NAS can discover competitive image classifiers when population turnover is carefully regularized [20, 21]. Population Based Training jointly adapts weights and hyperparameters in an asynchronous population [22]. Practical Bayesian optimization [26] and CMA-ES tutorials [24] inform local Bayesian and continuous evolutionary prior families. These are **used** for GA tournament/crossover branches, PBT-style exploit–mutate, and literature-combo priors, implemented at synthesizer-trial scale.

2.6 Evolutionary reinforcement learning hybrids

Khadka and Tumer's evolution-guided policy gradient (ERL) interleaves an evolutionary population with an RL actor to mitigate sparse reward and brittle hyperparameters in deep RL [23]. We **use** ERL as the spirit of interleaving GA population steps with PPO updates at tiny population sizes. We do not claim a full ERL robotics stack.

2.7 Mixture-of-experts soft gating

Sparsely gated MoE layers increase capacity via conditional computation [25]. We **use** MoE as motivation for soft gates over heterogeneous seam blocks (e.g., U-Net vs dilated vs attention cells) under residual scoring, at bake-scale widths.

2.8 Used versus screened

Used (methodological or comparative anchors).

- Discontinuity / VA lineage: [1–3].
- Unsupervised restoration philosophy: [4, 5, 29].
- Modular / differentiable audio DSP context: [6].
- Audio DL and compact waveform cells: [7–15, 30, 31].
- NAS / RL controllers / PPO: [16–19, 27, 28].
- Evolutionary NAS and population HPO: [20–22, 24, 26].
- ERL hybrid outer loop: [23].
- MoE soft gating prior: [25].

Screened (contrast only).

- **MAML** [32]: bi-level / fast-adaptation framing. We do not differentiate through a deep task network. Inner fits are coordinate-style updates on bake parameters. Outer rank is residual R.
- **DARTS** [33]: continuous relaxation of architecture search. We keep discrete ops + small cells under residual scoring.
- **Demucs / Hybrid Demucs / realtime waveform enhancement** [34–36]: strong separators and enhancers, but full multi-scale / hybrid STFT stacks are too heavy per meta trial under bake-cell search budgets.
- **DiffWave** [37]: diffusion vocoder. Screened as a generative sampling stack outside the seam bake operator set.
- **SEGAN-style adversarial enhancement** [38]: optional tiny adversarial cues only. Full GAN training loops screened for per-trial cost.

Positioning. Relative to this map, this work contributes (i) a prolonged residual objective for periodic seam repair, (ii) a hybrid RL+GA(+PBT) meta-outer loop with named branches, and (iii) a used/screened literature protocol with downloaded source PDFs. Claims stay inside cycle-local seam restoration under a declared compute budget. The primary claim is protocol-level: residual-scored meta-search for cycle-local seam operators.

3 Methods

3.1 Notation

Throughout, $x \in \mathbb{R}^L$ is the cracked single-period wavetable (engine input with open-wrap cliff). Let $\mathscr{H}$ be the space of bake-cell configurations (discrete operator graph plus continuous bake parameters). The bake operator is

$$\Theta : \mathbb{R}^L \times \mathscr{H} \to \mathbb{R}^L, \qquad y = \Theta(x; h), \quad h \in \mathscr{H}. \tag{1}$$

When continuous parameters alone are emphasized we write θ for the continuous slice of h and $y = \Theta(x; \theta)$. $r^\star \in \mathbb{R}^L$ is the procedural ideal sibling defined below. Tiling length N (typically N=16) forms $y_{\text{tiled}} = \text{Tile}(y, N)$ and $r^\star_{\text{tiled}} = \text{Tile}(r^\star, N)$. Seam width W (code: `SEAM_W`, typically 8) is the neighborhood edited by classical fade and polish ops. $R \in [0, 1]$ is the prolonged residual score (1 = best). These symbols follow a different convention from Noise2Noise clean/noisy pairs.

3.2 Problem and ideal sibling

Problem. Given a cracked period x, choose a bake configuration h so that the repaired cycle $y = \Theta(x; h)$ matches a no-cliff ideal under prolonged tiling.

Ideal sibling (Q1). Let $G(\text{seed}, \text{cliff})$ denote the procedural generator used by the CUDA FitCell batch maker (`make_batch` in `overnight_gpu_rl_arch.py`): one draw of frequency, phase, and mid-cycle content, with an optional open-wrap cliff of width W at both ends. For any seed,

$$r^\star = G(\text{seed}, \text{cliff=off}), \qquad x = G(\text{seed}, \text{cliff=on}). \tag{2}$$

Engine and ideal therefore share the mid-cycle content of one draw and differ by the seam cliff (plus any seam-boosted noise applied only on the engine path). The ideal is *not* a latent studio ground truth and is *not* a method output. For an arbitrary cracked period from this family, the sibling is obtained by replaying the same seed with the cliff withheld. That is exactly the pair returned by `make_batch`. This sibling relationship makes unsupervised ranking possible without paired studio recordings [4, 5].

Table 1: Core symbols used in Methods.

Symbol	Meaning
L	Cycle length (samples per period)
x	Cracked engine wavetable period (cliff on)
$y = \Theta(x; h)$	Baked cycle after seam ops / residual cell
Θ	Bake operator $\mathbb{R}^L \times \mathscr{H} \to \mathbb{R}^L$
$h \in \mathscr{H}$	Bake-cell configuration (ops + continuous θ)
$r^\star$	Ideal sibling (same seed; cliff withheld)
G	Procedural period generator (cliff on/off)
N	Prolong tiling count
W	Seam neighborhood width
R	Prolonged residual score in $[0, 1]$ (1 = best)

3.3 Outer objective and residual score R

Single outer objective. Every candidate bake (classical controls and searched cells alike) is ranked by maximizing absolute prolonged residual against the same ideal sibling:

$$\max_{h \in \mathscr{H}} R\big(y, r^\star\big) \quad \text{subject to} \quad y = \Theta(x; h). \tag{3}$$

Equivalently, FitCell minimizes loss $1 - R$. Selection, population ranking, and champion updates use absolute R only.

Every scored row (no-bake, DualCosine, FIR, poly, VA residuals, neural favorite) applies some Θ to the cracked engine x, then compares tiled y to the *same* $r^\star$ under (4). The ideal sibling is the scoring target. It is not a method and not the no-bake baseline.

With tiling length N,

$$R = \text{clamp}\left(1 - \frac{\text{rms}(y_{\text{tiled}} - r^\star_{\text{tiled}})}{\max(\text{rms}(r^\star_{\text{tiled}}), \varepsilon)}, 0, 1\right), \tag{4}$$

where $\text{rms}(z) = \sqrt{\frac{1}{|z|}\sum_i z_i^2}$ and $\varepsilon > 0$ avoids division by zero. By construction $R \in [0, 1]$, R=1 when $y_{\text{tiled}} = r^\star_{\text{tiled}}$, and (when pre-clamp scores lie in $(0, 1)$) R is strictly decreasing in residual RMS for fixed ideal RMS. A method "beats" another iff its R is higher on the same $(x, r^\star)$ draw. Optional soft shape gates (when enabled) multiply R by factors that down-weight mid-cycle damage relative to seam-local error. We do not claim convergence guarantees for the hybrid outer loop, nor a general wrap-closure theorem for arbitrary bake cells. Outer-loop cost scales as population size $\times$ proposals per iteration $\times$ fit steps $\times$ batch forward cost (order 10^3–10^4 arch evaluations at the reported 5k gate).

Remark 1 (Noise2Noise motivation). Ranking without studio-clean pairs is possible when the ideal is a procedural sibling of the engine draw [4, 5].

3.4 Bake operator Θ and classical baselines

A **bake cell** $\Theta(\cdot;h)$ composes classical seam ops (DualCosine, polish, soft seam, FIR blends) with optional tiny residual cells (MLP/FIR/U-Net-lite/attention-lite/LSTM-lite) and optional MoE soft gates over experts (Appendix A, Algorithms 2–4). Bake parameters live in a bounded box; architecture strings are discrete graphs over those ops. Separate classical controls apply cycle-local BLIT/BLEP, PolyBLEP, and BLAMP residual corrections at the wrap on the cracked engine tile (same interface as DualCosine / FIR; not a full oscillator rewrite). Plain-language roles: BLIT/BLEP and PolyBLEP correct a *value* step at wrap; BLAMP corrects a *slope*/corner jump; DualCosine fades endpoints without a BLEP residual. They are scored in Section 5.7.

Four roles (do not conflate).

- **Ideal sibling** $r^\star$: cliff withheld; scoring target only (panel (a) in Figure 1).
- **No-bake** (passthrough): $\Theta_{\text{no-bake}}(x) = x$; unrepaired cracked engine scored vs $r^\star$ (panel (b)). Not $r^\star$, not a residual-net skip, not a learned identity map.
- **DualCosine**: classical raised-cosine end fade on x (panel (c)); one classical board row and, separately, a PPO advantage baseline (Remark 2).
- **Ours**: searched bake $y = \Theta(x;h^\star)$ from the hybrid outer loop (panel (d)).

On mild cliffs residual mass $\ll$ ideal RMS under (4), so no-bake R can sit near 0.97 while the wrap click remains audible. That is a "do nothing" reference, not a perceptual wrap score. Released JSON may still use the legacy key `identity` for no-bake; manuscript prose uses *no-bake*.

Classical (non-AI) comparison board. Learned / searched methods are always ranked against the full classical board on absolute R (same $r^\star$): no-bake, DualCosine / cosine fade, `seam_fir3`, classic quadratic, hann/crossfade/soft fades, detrend/ensemble, poly seam, and cycle-local BLIT/BLEP / PolyBLEP / BLAMP where scored. When tables show ΔR vs DualCosine, that is a reporting gap vs one classical bake, not the definition of reward.

Remark 2 (DualCosine centering is PPO advantage only (Q2)). The outer objective remains (3): maximize absolute $R(y, r^\star)$. In the PPO branch only, we optionally form a centered advantage

$$\rho = R - R_{\text{DualCosine}}, \tag{5}$$

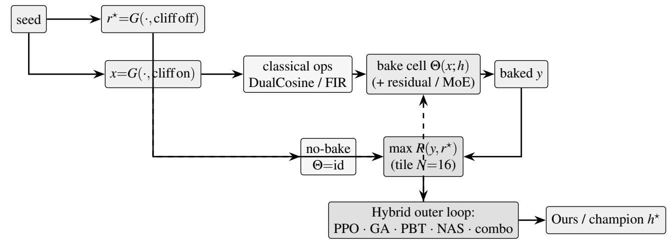

Figure 2: DenoiseOpt overview (before component deep-dives). Same-seed generator G yields ideal sibling $r^\star$ (cliff off) and cracked engine x (cliff on). Bake cell $\Theta(x;h)$ may include DualCosine/FIR and optional residual/MoE experts. No-bake is the identity control $x \mapsto x$. Outer objective is maximize absolute prolonged $R(y, r^\star)$. DualCosine centering (not shown) is PPO advantage shaping only. Dashed feedback: architecture proposals update the cell. Pseudocode in Appendix A.

so early-search advantages are roughly zero-mean. This does *not* change selection, FitCell loss, champion ranking, or the matched 5k meta-approach objective. All of those still maximize absolute R. DualCosine centering is soft relative shaping for the actor–critic, not a hard absolute ranking target.

3.5 Pipeline overview

Figure 2 summarizes the residual-scored hybrid loop before component detail. Each outer iteration proposes a bake configuration h, runs FitCell (Appendix A, Algorithm 5) to fit continuous parameters under loss $1 - R$, evaluates prolonged R vs $r^\star$, and updates the population / PPO policy. Pseudocode for GA, PPO, PBT, and the full hybrid rotation is in Algorithms 6–9.

3.6 Search space and FitCell

Each genome encodes: ordered op list (fade, polish, FIR, pin, residual write), depth, residual cell type (none, MLP, FIR, U-Net-lite, attn-lite, LSTM-lite), optional MoE expert count $K \in \{2, 3, 4\}$, and continuous θ in the CUDA runner box bounds. The searchable bake vocabulary includes a width-capped 1–2 layer LSTM block over the seam window or full cycle (distinct from the fixed supervised seq-LSTM SOTA baseline). The live hybrid tag is `PPO+GA+PBT+NAS+depth+MoE` with LSTM enabled in the block graph.

FitCell draws sibling batches via G, applies $\Theta(\cdot;h)$, and steps Adam on $1 - R$ with early stopping on plateau (Algorithm 5).

3.7 Hybrid outer loop

Each iteration rotates among branches: PPO, GA, PBT, NAS, and combo. Selection and FitCell are always by absolute residual R (maximize toward the ideal sibling; loss $1 - R$).

Near-ceiling R and why reward / search HPs matter. On the sine+cliff generator, no-bake already sits near $R \approx 0.97$, and strong classical / searched bakes live in a narrow band toward 1 (often 0.99–0.991). Absolute gaps of order 10^{-3}–10^{-2} are scientifically real but tiny as raw

PPO/GA signals. Without shaping, advantages collapse. Hyperparameters (clip, entropy, mutation rate, population size, fit steps, learning rate) then dominate whether the outer loop can climb the last percent. We therefore treat **reward shaping and search hyperparameters as first-class tunables** alongside bake architecture. In the present hybrid, PPO uses a searchable shaping mode among {`abs_r`, `vs_dualcosine`, `vs_nobake`, `neglog_gap`} (PBT-mutated; default $\rho = R - R_{\text{DualCosine}}$ per Remark 2). PBT mutates fit / PPO hyperparameters and reward mode in-population; plateau adaptation retunes branch mix and depth/MoE bias when champions stall. The matched 5k meta-approach comparison (Section 3.8) is the sole GPU experimentation vehicle for outer-loop family, HP, and reward-shaping sweeps under a shared FitCell budget.

3.8 Meta-learning approach comparison

Beyond the hybrid outer loop, we compare five additional outer-loop families under a matched 5,000-evaluation budget, shared search seed `1902771841`, and the same FitCell / prolonged-R protocol (LSTM and xLSTM included in the searchable bake vocabulary). Table and figure legends use the short display names below (artifact folders keep stable code keys; see `NOMENCLATURE.md`):

- **Random NAS** (`random`): independent uniform draws over bake graphs and fit hyperparameters (control).
- **Cont. CMA-ES** (`cmaes`): diagonal covariance-matrix adaptation over a continuous encoding of depth, width, cell kind, block logits, and fit hyperparameters [24].
- **Arch REINFORCE** (`reinforce`): vanilla policy gradient over discrete architecture-edit actions (no clipped surrogate; contrast to PPO [17, 19]).
- **Aging evolution** (`aging_evo`): regularized evolution with oldest-member replacement after tournament parent selection [20, 21], distinct from the hybrid's non-aging tournament GA branch.
- **TPE Bayes NAS** (`tpe`): lightweight tree-structured Parzen estimator over discrete bake categoricals in the spirit of practical Bayesian optimization [26].

We also run **Ours (hybrid GA–PPO)** (`hybrid_lstm`): the multi-branch hybrid loop for 5,000 evaluations with LSTM/xLSTM in the block vocabulary and online reward-mode / fit–PPO hyperparameter co-tuning, as the proposed matched-budget arm. Results appear in Table 13 and Figures 11–12; healed-sample overlays in Figure 13.

Remark 3 (Trial complexity). Per outer iteration the dominant cost is $O(T_{\text{fit}} \cdot C_{\text{fwd}})$ for fit steps times one forward bake. Bake-width caps (tiny residual cells) keep C_{fwd} far below full-band Demucs/DiffWave forwards screened in Related Work.

Table 2: Hybrid search hyperparameters (RTX 3090 DenoiseOpt CUDA search runner / `denoise_meta_evo.py` defaults). Co-tuned with reward shaping under near-ceiling R (Section 3.7).

Parameter	Value
Population size n	12
GA tournament k	3
GA crossover fraction p_c	0.5 (else clone elite parent)
GA mutation	≥ 1 mutate after inherit (+50% second mutate)
PPO clip ε	0.2 (default; searchable $\{0.1, 0.2, 0.25, 0.3\}$)
Adam learning rate (fit)	3×10^{-3} (default; PBT-searchable)
Adam learning rate (policy)	3×10^{-4}
Entropy coefficient (PPO)	0.02 (default; searchable $\approx$ 0.005–0.06)
PBT exploit / mutate	elite fraction 0.25; multi-step arch+hp mutate
MoE routing	`sequential` or `moe_parallel` when enabled
Plateau boredom threshold	1000 iters without champ improve
Cycle length L / tiles N / seam W	256 / 16 / 8
Fit steps / batch (default)	24 / 48
Algo tag	`PPO+GA+PBT+NAS+depth+MoE`

3.9 Hyperparameters

Table 2 lists the CUDA search defaults used for the 5k-gate freeze reported in Results. Under the near-ceiling residual regime (Section 3.7), these outer-loop and fit hyperparameters are co-tuned with architecture; PBT and plateau adaptation continue to mutate several of them online. Companion listings: Algorithms 1–9 (Appendix A) and `docs/PSEUDOCODE.md`.

4 Experiments

Protocol freeze. Evaluation follows `EVAL_PROTOCOL.md` in this paper folder (extended for peer-review strata). Primary metric: prolonged R. Secondary: SNR, SDR, wrap-jump (diagnostic), and required seam-local edge RMSE on tiled audio. Holdout seed `20260719`. Search seed `1902771841`. We report the 5k-gate freeze plus frozen holdout. We make no unfinished-budget claims. PESQ/STOI stay out of scope on non-speech cycles. LibriSpeech and MUSDB are not used. Venue positioning: DSP / synthesis artifact repair (DAFx / AES / arXiv cs.SD). This is not a general speech-enhancement leaderboard paper.

4.1 Dataset

Creation. The paper-facing evaluation corpus is a frozen draw from the same synthetic generator used by the CUDA search runner (`make_batch` in `overnight_gpu_rl_arch.py`). Each cycle has length L=256. A two-harmonic sine is drawn with frequency

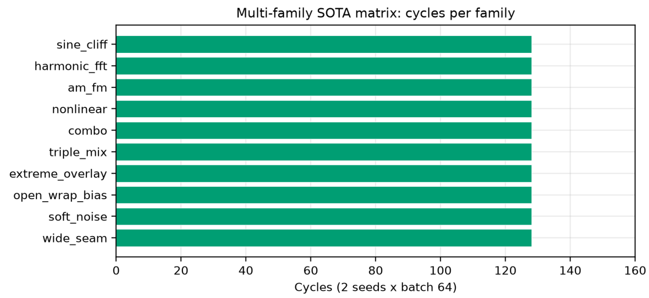

Figure 3: Multi-family SOTA board: 128 cycles per family (2 seeds $\times$ batch 64).

$\sim \mathscr{U}(1,4)$ and phase $\sim \mathscr{U}(0,2\pi)$. An open-wrap seam cliff of amplitude $\pm\mathscr{U}(0.08,0.43)$ is applied over `SEAM_W=8` samples at both ends. Seam-boosted Gaussian noise (scale 0.02 at the ends, 0.15$\times$ in the mid-cycle) is added to form the engine input. The ideal withholds the cliff and is used only as the scoring target $r^\star$ (never as a method output). No-bake leaves the cracked engine unchanged and scores that unrepaired x against the same $r^\star$. Prolonged residual R tiles each cycle N=16 times before the RMS ratio. The holdout seed is `20260719` (distinct from the hybrid search seed `1902771841`). The search campaign draws fresh i.i.d. batches from this generator and does not train on the frozen holdout tensors. There is no supervised clean/noisy training split of studio audio.

Multi-family diversity ($\geq$ 20 **waveforms).** Beyond the single sine+cliff holdout, Phase 3a scores methods on **20** generative draws: ten family variants (`sine_cliff`, `harmonic_fft`, `am_fm`, `nonlinear`, `combo`, `triple_mix`, `extreme_overlay`, `open_wrap_bias`, `soft_noise`, `wide_seam`) $\times$ two seeds (`20260719` and `20260720`), batch 64 each. Figure 3 counts cycles per family (128 each). These Python generative variants stress harmonic / AM–FM / nonlinear / overlay / open-wrap behavior of `make_batch` and remain the primary SOTA matrix. Separately, `export_sound_bench_tiles` exports **20** Rust `sound_bench` engine/ideal pairs (10 families $\times$ 2 seeds, L=256, byte-aligned with `generate_sound` / `generate_sound_ideal`) scored under the same residual geometry as a secondary transfer block (Table 14). Every method sees the same tensors per waveform. **Primary classical comparison** is absolute prolonged R against the non-AI board (no-bake, DualCosine, `seam_fir3`, poly, fades, VA residuals; Section 3.4). The search objective is maximize R toward the ideal sibling (best R=1). DualCosine is one classical comparator and a PPO centering constant. It is not the reward target.

Dataset statistics. Table 3 and Figures 4–5 inventory every evaluation board used in this paper. The primary procedural holdout is a frozen 4,096-cycle metrics draw (L=256, N=16 tiles, seed `20260719`) plus a matched score batch of 64 cycles (five consecutive seeds for classical mean±std).

Table 3: Dataset statistics (evaluation boards). L=256 unless noted. Hours are raw source duration on disk where measurable, not tiled period count.

Board	Size	Notes
Holdout metrics	4,096 cycles	seed `20260719`; N=16 tiles
Score batch	64 cycles	classical / favorite tables
Multi-family SOTA	1,280 cycles	10 families $\times$ 2 seeds $\times$ 64
Factory export	25 periods	ReelSynth mid-morph frames
AKWF OA	24 periods	CC0 single-cycle WAVs
Meta 5k compare	6 approaches	shared seed; batch 48
CWRU	256 periods	4 MAT files; DE @12 kHz; $\approx$0.025 h raw
MFPT	128 periods	shaft-rate windows
MIT-BIH	256 periods	10 records @360 Hz; $\approx$3.76 h raw
PTB-XL	256 periods	94 `*_hr` @500 Hz; $\approx$0.54 h raw
synth CNC / PMU	256+256	procedural pilots

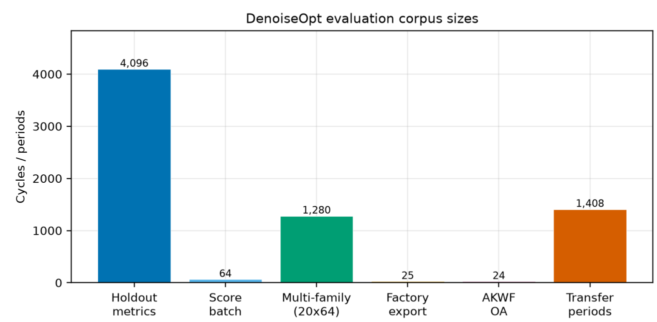

Figure 4: Corpus sizes across evaluation boards (cycles or periods). Holdout metrics draw, matched score batch, multi-family SOTA, factory/AKWF wavetable-native sets, and summed transfer periods.

Multi-family SOTA scores 20 waveforms (10 families $\times$ 2 seeds, batch 64; 1,280 cycles). Wavetable-native boards use ReelSynth factory exports (n=25) and AKWF CC0 cycles (n=24). Sci/eng transfer tiles 1,408 periods across CWRU, MFPT, MIT-BIH, PTB-XL, and synthetic CNC/PMU (cache metas under `signal_heal_transfer/cache/`). Search never trains on the frozen holdout; N2N/seq baselines use disjoint train seeds `424242`/`424243`. Artifacts: `dataset_inventory.json`, `fig_dataset_*.png`.

Dataset metrics. A metrics draw of 4,096 cycles under the holdout seed yields mean ideal RMS 0.721 ± 0.016, mean engine residual RMS 0.021 ± 0.013, and mean absolute wrap jump $|x_0 - x_{L-1}|$ of 0.913 ± 0.634. No-bake (passthrough) on that draw scores mean $R \approx 0.971$. Method tables use a matched score

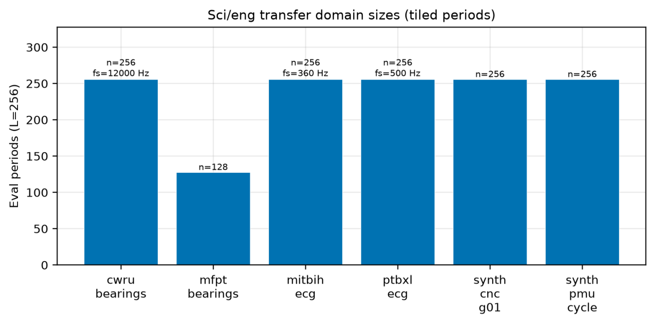

Figure 5: Sci/eng transfer domain sizes: tiled eval periods at L=256 (with source f_s where applicable).

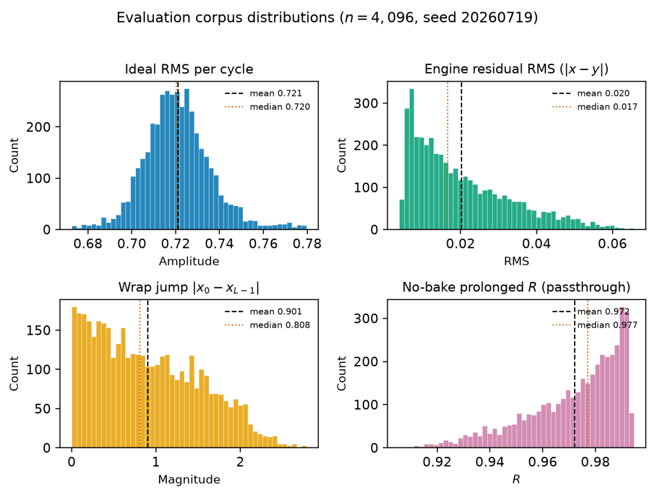

Figure 6: Holdout distribution statistics (seed `20260719`, n=4,096 cycles). Distinguish: ideal sibling $r^\star$ (cliff withheld); no-bake = unrepaired cracked engine (not $r^\star$); DualCosine = one classical bake. Top left: ideal RMS per cycle. Top right: engine residual RMS before any bake. Bottom left: wrap jump magnitude $|x_0 - x_{L-1}|$ (cliff hardness proxy). Bottom right: no-bake prolonged R (passthrough, $x \mapsto x$). Dashed and dotted vertical lines mark mean and median. Hard-cliff reporting (Section 5) uses the top 10% wrap-jump stratum where no-bake R drops.

batch of 64 cycles from the same seed (tensors under `brand/artifacts/canonical_eval_dataset/`), five consecutive seeds for mean±std on the classical table, and the 20-waveform matrix for SOTA secondary metrics. Figure 6 summarizes the 4,096-cycle metrics draw: ideal RMS clusters near 0.72, wrap jumps span nearly three orders of magnitude (median ≈ 0.83), and no-bake R is high on easy tiles but falls below 0.91 on the hardest decile of wrap jumps. Figure 1 is the documented sine-wrap edge case: a high-$|wrap|$ holdout tile where cliff modulation is visible, scored with DualCosine and the top neural favorite under the same residual R (tile $R \approx 0.9917$, favorite batch mean $R \approx 0.991$).

Hybrid search campaign (5k clean gate). A CUDA search tagged `PPO+GA+PBT+NAS+depth+MoE` runs with a large declared iteration budget (order 10^5). At the reported freeze (run `gpu-rl-arch-20260719T083019Z`, ≈6,922 clean iters / ≈ 7.9 h on RTX 3090) the champion residual is $R \approx 0.99093$ against DualCosine baseline ≈ 0.81658 ($\Delta R \approx +0.174$). Branch-best residuals at the freeze: GA ≈ 0.99016, PBT ≈ 0.99012, PPO ≈ 0.98924, combo ≈ 0.98854, NAS ≈ 0.98677. Isolated short-budget re-runs (GA-only, PPO-only, GA+PPO, full, 150 iters, same search seed, plateau adapt off) are reported beside those freezes in Table 11. Figures in Section 5 regenerate from `history.jsonl` near this gate.

Inference, classical, and learned baselines. High-R fitted cells ($R \geq 0.98$) are timed on CUDA (batch 64, prolonged residual). Ten non-learnable bake denoisers share the frozen canonical holdout against the neural favorite (Table 5). Cycle-local BLIT/BLEP, PolyBLEP, and BLAMP seam repairs are scored on the same holdout / cliff / multi-family protocol (Table 10; `va_seam_blep.json`) and illustrated on the intro high-wrap tile (Figure 10). Fixed-arch MLP-on-R and CNN/U-Net-lite-on-R baselines (no outer NAS) are trained on $1-R$ with the same generator and scored in the SOTA matrix. Phase B adds a primary corrupt→corrupt Noise2Noise seam CNN (train seed `424242`, disjoint from holdout), a secondary sibling-supervised ceiling on the same architecture, and lightweight LSTM / depthwise 1D-CNN seq restorers (train seed `424243`). No search-champion weights initialize those baselines. Family-conditional hardness uses the separate Rust procedural 100,000-cycle bench reported in Results.

Cliff strata and wavetable-native realism. A 4,096-tile holdout draw (seed `20260719`) is stratified by engine wrap-jump percentiles (top 25% / top 10%; cutoffs frozen in `cliff_strata.json`). Hard-cliff is the operative regime for claims about DualCosine collapse vs favorite retention. Edge RMSE is the locked required seam-local secondary; click energy is optional. Wavetable-native transfer (Phase F1) scores true ReelSynth-exported factory periods (primary) and external AKWF CC0 WAVs (secondary) under the same close-seam→open-wrap protocol. Procedural stand-ins are tertiary smoke only. Speech/music LibriSpeech/MUSDB corpora remain off.

Audible demos and vibrato spectrograms. Downloadable wavetable clips (no-bake / DualCosine / Ours) live under `reelsynth/brand/artifacts/meta_approach_compare/he` (44.1 kHz mono, A4, 3 s; rebuild: `scripts/export_meta_hear_samples.py`). A slow-vibrato spectrogram protocol (`scripts/bench_vibrato_spectrogram.py`) scores the same bake cells under modulated-pitch playback; figures appear in Section 5.12. These assets support informal audition and spectrogram inspection. Formal MOS/MUSHRA scores were not collected; the planned formal listening protocol is stated in Section 5.16.

4.2 Sci/eng wrap-heal transfer protocol

Beyond wavetable-native export/AKWF benches, we transfer the same residual protocol and hybrid outer loop (`hybrid_lstm`) to cycle-local wrap tasks outside musical wavetables. Period length is fixed at L=256. The outer objective remains maximize prolonged R vs an ideal sibling. Pilot budget: 150–250 outer iterations per dataset, `fit_steps` = 40 (hours-scale, not multi-day). Artifacts: `reelsynth/brand/artifacts/signal_heal_transfer/`.

Datasets that ran.

- **CWRU bearings** (real): DE @12 kHz; per-rev windows via RPM; ideal = cubic resample; engine = linear (bad-COT proxy) + wrap cliff.
- **MFPT bearings** (real): fixed 25 Hz shaft; same cubic/linear+cliff construction (Figshare mirror of MFPT Fault Data Sets).
- **MIT-BIH ECG** (real): R–R normal beats; ideal = local mean template + mild endpoint equalize; engine = beat + cliff.
- **PTB-XL ECG** (real subset): PhysioNet `records500` (`*_hr`, 500 Hz) lead-I beat windows; same classical ECG construction. This is a subset pilot (n=256 periods from 94 records), not the full PTB-XL corpus.
- **Synthetic CNC G01** (`synth_cnc_g01`): closed-contour residual with sharp corners vs wide G2-ish blend (KIT CNC DOI login-walled).
- **Synthetic PMU/AC cycle** (`synth_pmu_cycle`): harmonic voltage cycle with wrap cliff (IEEE DataPort PMU OA account-walled).

Honesty labels (required). All transfer tables are *classical-board* comparisons (no-bake, DualCosine, fades, endpoint pin, seam FIR, plus domain classical COT / SBMM-lite / spline proxies). We do *not* claim deep SOTA on diagnosis or clinical enhancement. Rows marked *not executed*: Paderborn KAt deep campaign (CC BY-NC + multi-GB), Cycle-GAN / BeatDiff under this residual protocol, full PTB-XL 500 Hz corpus, real KIT CNC / IEEE 39-bus PMU recordings, and formal MOS/MUSHRA listening. Synthetic CNC/PMU rows are protocol pilots, not claims on industrial toolpaths or live PMU streams. Results appear in Section 5.15.

5 Results

Family-conditional hardness (completed bench). RQ: which procedural families dominate remaining residual mass? On the fitted DenoiseOpt 100,000-cycle bench, denoise score D is lowest for `nonlinear` ($D \approx 0.39$) and `combo` ($D \approx 0.40$), then `triple_mix` / `extreme_overlay` ($D \approx 0.51$–0.52), while `harmonic_fft` and `am_fm` sit near $D \approx 0.69$–0.70. Lit-combo champion family stress (prolonged residual) similarly ranks `open_wrap_bias` ($R \approx 0.83$) and `extreme_overlay` / `combo` ($R \approx 0.88$–0.89) below `harmonic_fft` ($R \approx 0.95$). Hard sounds recur.

Table 4: Top-5 distinct fitted architectures on the CUDA inference bench (batch 64, prolonged R, seed geometry matching the search campaign). Favorite is latency-best inside the near-max-R band ($R \geq R_{\max} - 5 \cdot 10^{-4}$).

Tag	R_{saved}	R_{live}	ms/batch	params
`iter_000157`	0.9914	0.9913	7.18	104 484
`iter_000205`	0.9911	0.9907	7.29	70 562
`iter_000235`*	0.9910	0.9915	3.15	37 489
`iter_000397`	0.9910	0.9907	6.80	97 262
`iter_000229`	0.9909	0.9902	6.67	100 500

*Favorite (fastest in $R \geq R_{\max} - 5 \cdot 10^{-4}$ band).

Hybrid search (5k-gate freeze). RQ: what champion R does the hybrid search campaign reach at the clean 5k-iteration gate? At the reported 5k-gate search freeze ($\approx$ 6,922 clean iterations, $\approx$ 7.9 h on RTX 3090) the champion residual is $R \approx 0.99093$ with DualCosine ≈ 0.81658 on the CUDA search runner ($\Delta R \approx +0.174$). Monitoring plots below are regenerated from that freeze. Declared larger budgets continue. This paper reports the gate freeze only.

Inference time at matched score. RQ: among near-max-R fitted cells, which is fastest at matched score? Across diverse high-R fitted cells ($R \geq 0.98$), we timed CUDA forward passes (batch 64, prolonged residual). Within a tight band of the best residual ($R \geq R_{\max} - 5 \cdot 10^{-4}$), the **favorite** is the lowest-latency model: `champion_iter_000235` with $R \approx 0.9910$, $\approx$ 3.15 ms/batch, $\approx$ 37k parameters (versus a higher-capacity max-R cell at $R \approx 0.9914$ but $\approx$ 7.18 ms/batch / $\approx$ 104k params). Selection rule: maximize R, then minimize latency inside the band.

5.1 Top-5 architectures

RQ: which fitted bake graphs achieve near-max R at lowest latency on the inference bench? Table 4 ranks five distinct fitted bake graphs from the CUDA inference bench (`inference_bench.json`, batch 64), preferring residual R first and latency on ties, and skipping near-duplicate copies of the same topology signature.

Classical vs favorite (canonical holdout). Table 5 scores the classical (non-AI) bake set (no-bake, DualCosine, and `seam_fir3`) plus the neural favorite on the frozen corpus of Section 4.1 (CUDA, batch 64, seed `20260719`). Every row uses the same ideal sibling $r^{\star}$; higher R means closer to that sibling. DualCosine reaches $R \approx 0.825$ ($\approx$ 1.35 ms/batch), near the search-monitor DualCosine ≈ 0.817 on live draws from the same generator. The best *active* classical repair is seam-local 3-tap FIR (`seam_fir3`) at $R \approx 0.961$ / $\approx$ 0.67 ms/batch (0.962 ± 0.002 over five seeds). No-bake (passthrough; unrepaired engine vs $r^{\star}$) scores $R \approx 0.972$. It is a do-nothing reference, not the ideal. The neural favorite reaches $R \approx 0.991$ (0.991 ± 0.0002 over five seeds) at

Table 5: Method scores on the canonical sine+cliff holdout (batch 64, seed `20260719`, L=256, N=16). Mean±std over five consecutive seeds. Every method is scored vs the same ideal sibling $r^{\star}$. No-bake = unrepaired engine (not $r^{\star}$). Rank by absolute R against the classical board; DualCosine is one classical row, not the sole comparator. Primary metric is prolonged R (1 = best).

Method	R	R (5-seed)	ms/batch
neural favorite	0.9911	0.9912 ± 0.0002	2.56
no-bake (passthrough)	0.9718	0.9726 ± 0.0021	0.31
`seam_fir3`	0.9610	0.9616 ± 0.0019	0.67
classic quadratic	0.8449	0.8453 ± 0.0117	1.59
DualCosine	0.8249	0.8258 ± 0.0137	1.35
cosine fade	0.8249	0.8258 ± 0.0137	1.59
crossfade	0.8116	0.8130 ± 0.0145	1.50
hann blend	0.8048	0.8060 ± 0.0156	0.54
soft wide fade	0.7432	0.7378 ± 0.0171	2.64
detrend	0.4079	0.4091 ± 0.0386	0.36
ensemble detrend+DC+FIR	0.4024	0.4027 ± 0.0387	1.56

≈ 2.56 ms/batch (≈ 37k params): $\Delta R \approx +0.019$ vs no-bake, $\Delta R \approx +0.030$ vs `seam_fir3`, and $\Delta R \approx +0.166$ vs DualCosine.

5.2 SOTA matrix (multi-family + secondary metrics)

RQ: does the meta favorite beat the classical non-AI board (no-bake, DualCosine, FIR, poly, fades) and fixed MLP/CNN baselines across twenty generative families? Table 6 reports mean±std over the 20-waveform multi-family set (Section 4.1), with canonical-holdout SNR/SDR/wrap-jump for the same method set from `sota_matrix.json`. Figures 7–8 plot the same ranking. PESQ/STOI/MUSHRA are omitted for non-speech cycles (Limitations). Primary reading is absolute R rank vs classical rows (no-bake 0.965, poly 0.966, `seam_fir3` 0.955, DualCosine 0.814). Fixed MLP-on-R and CNN/U-Net-lite-on-R (no outer NAS) sit between DualCosine and the meta favorite. Paired Wilcoxon signed-rank (normal approx.) of favorite vs DualCosine on the 20 waveforms: W=0, $p \approx 8.9 \times 10^{-5}$, mean ΔR=+0.162 (bootstrap 95% CI $[0.155, 0.170]$); vs no-bake the multifamily mean gap is smaller (≈ +0.012) but still favors the searched bake on this generator.

5.3 Hard-cliff strata

RQ: does high no-bake R hide DualCosine failure on large wrap jumps? Table 7 stratifies a 4,096-tile holdout draw (seed `20260719`) by engine wrap-jump percentiles (p75≈ 1.387, p90≈ 1.832; frozen in `cliff_strata.json`). No-bake R barely moves (all 0.9715 → top-10% 0.9667) because residual mass remains small relative to ideal RMS under (4). DualCosine collapses on hard cliffs (0.819 → 0.657), while the meta favorite holds ($R \approx 0.9915$). Favorite wrap-jump on the top-10% subset stays ≈ 1.83 (vs no-bake 2.09). The claim is tiled residual repair, not endpoint-zeroing. **Edge RMSE** is the

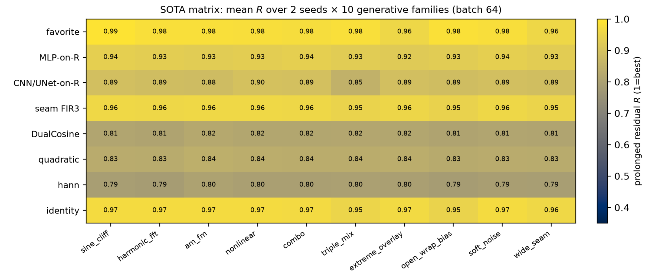

Figure 7: Colorblind-safe (`cividis`) heatmap of mean prolonged R for selected methods × generative families (2 seeds each). Higher is better.

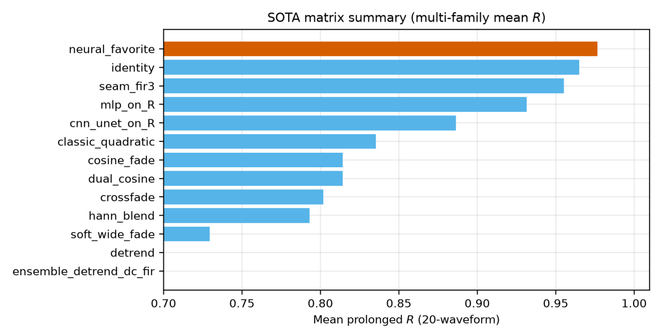

Figure 8: Multi-family mean prolonged R by method (same 20-waveform board as Table 6).

locked required seam-local secondary (EVAL_PROTOCOL): on the top-10% subset it drops from no-bake 0.095 to favorite 0.021 (DualCosine 0.990). Click energy remains an optional diagnostic.

5.4 N2N and seq baselines

RQ: how does DenoiseOpt compare to true Noise2Noise-style and lightweight seq restorers on the same generator? Primary corrupt→corrupt N2N (20k Adam steps, 53k params, train seed `424242`) reaches canonical $R \approx 0.9917$, matching the meta favorite within ≈ 0.001. Secondary sibling-supervised N2N reaches $R \approx 0.9933$. A bidirectional LSTM (76k params, 15k steps, train seed `424243`) is the supervised ceiling at $R \approx 0.9952$. A tiny depthwise 1D CNN (3.2k params) reaches $R \approx 0.9864$. Holdout tiles never enter training. Classical DualCosine / `seam_fir3` remain in Table 7. Figure 9 plots these canonical holdout scores beside no-bake, DualCosine, and the neural favorite.

5.5 Wavetable-native transfer

RQ: does the wrap protocol transfer beyond synthetic sine cliffs onto *true* exports and third-party OA cycles? Table 9 scores ReelSynth-exported factory periods (primary realism, n=25; byte-true factory frames resampled to L=256) and Adventure Kid Waveforms (AKWF) CC0 single-cycle WAVs (secondary, n=24) after close-seam idealization and the same open-wrap cliff. Procedural stand-ins are demoted to tertiary smoke and no longer carry the primary realism claim. On

Table 6: SOTA comparison on 20 generative waveforms (10 families $\times$ 2 seeds, batch 64). R/SNR/SDR/wrap-jump are multi-family mean$\pm$std. Rank by absolute R vs the classical non-AI board (no-bake, poly, FIR, DualCosine, ...). Column ΔR^* is vs DualCosine only (one classical gap for reporting; canonical freeze seed `20260719`); ms* likewise. The search objective is maximize absolute R toward the ideal sibling, not DualCosine matching. Modern baselines (N2N/seq/poly) use disjoint train seeds (`424242`/`424243`); poly is classical (no train). Primary metric R (1 = best; closer to ideal sibling). No PESQ/STOI/MUSHRA.

Method	R (20-wave)	SNR (dB)	SDR (dB)	wrap-jump	ms*	params	ΔR^*
neural favorite	0.977 ± 0.010	35.0 ± 3.2	35.1 ± 3.2	0.90 ± 0.14	2.44	37 489	+0.166
N2N corrupt→corrupt	0.970 ± 0.021	33.9 ± 4.1	34.0 ± 4.1	0.90 ± 0.15	–	53 506	+0.167
seq CNN1D	0.969 ± 0.016	32.2 ± 3.5	32.4 ± 3.5	0.92 ± 0.15	–	3 242	+0.162
N2N sibling-sup.	0.969 ± 0.022	33.7 ± 4.3	33.8 ± 4.3	0.91 ± 0.15	–	53 506	+0.168
poly seam (d=3)	0.966 ± 0.008	31.2 ± 2.1	31.3 ± 2.1	0.91 ± 0.15	1.47	0	+0.148
no-bake	0.965 ± 0.008	30.9 ± 1.9	31.0 ± 1.9	0.91 ± 0.15	0.00	0	+0.147
seq LSTM	0.956 ± 0.037	32.2 ± 5.8	32.3 ± 5.8	0.86 ± 0.15	–	75 746	+0.170
`seam_fir3`	0.955 ± 0.005	28.1 ± 1.2	28.1 ± 1.2	0.46 ± 0.08	0.59	0	+0.136
MLP-on-R	0.932 ± 0.006	24.6 ± 0.9	24.6 ± 0.9	0.64 ± 0.07	2.76	8 604	+0.113
CNN/U-Net-on-R	0.886 ± 0.012	19.5 ± 0.9	19.5 ± 0.9	0.48 ± 0.06	5.01	11 752	+0.069
classic quadratic	0.835 ± 0.012	17.9 ± 1.1	17.7 ± 1.2	0.32 ± 0.05	1.78	0	+0.020
DualCosine	0.814 ± 0.013	16.9 ± 1.2	16.8 ± 1.2	0.32 ± 0.05	1.48	0	0
cosine fade	0.814 ± 0.013	16.9 ± 1.2	16.8 ± 1.2	0.32 ± 0.05	2.64	0	0
crossfade	0.802 ± 0.013	16.7 ± 1.2	16.4 ± 1.2	0.91 ± 0.15	1.70	0	−0.013
hann blend	0.793 ± 0.015	16.1 ± 1.2	15.8 ± 1.2	0.32 ± 0.05	0.57	0	−0.020
soft wide fade	0.729 ± 0.016	14.0 ± 1.1	13.6 ± 1.1	0.63 ± 0.08	3.69	0	−0.082
detrend	0.375 ± 0.040	5.5 ± 1.0	5.4 ± 1.0	0.00 ± 0.00	0.12	0	−0.417
ensemble DC+FIR	0.368 ± 0.040	5.1 ± 1.0	5.0 ± 1.0	0.12 ± 0.02	1.99	0	−0.422

Table 7: Cliff strata on 4,096 holdout tiles (seed `20260719`). Strata by engine wrap-jump. Primary R (1 = best) and mean wrap-jump after bake. N2N primary = corrupt→corrupt; N2N secondary = sibling-supervised ceiling; seq = engine→ideal.

Method	all R	all jump	top25 R	top25 jump	top10 R	top10 jump
seq LSTM (ceiling)	0.9953	0.865	0.9955	1.637	0.9955	1.810
N2N sibling-sup.	0.9936	0.879	0.9936	1.668	0.9934	1.842
N2N corrupt→corrupt	0.9922	0.870	0.9924	1.647	0.9922	1.821
neural favorite	0.9911	0.869	0.9915	1.645	0.9915	1.825
seq CNN1D	0.9863	0.881	0.9869	1.684	0.9867	1.871
no-bake	0.9715	0.913	0.9713	1.797	0.9667	2.094
`seam_fir3`	0.9611	0.456	0.9461	0.893	0.9429	1.034
DualCosine	0.8193	0.335	0.6864	0.290	0.6574	0.335

n: all 4096, top25 1024, top10 410.

exported factory periods the favorite ($R \approx 0.911$) trails no-bake / `seam_fir3` / poly (domain shift) while still beating DualCosine (0.883). On AKWF OA cycles the favorite ($R \approx 0.970$) matches no-bake and beats DualCosine (0.936). LibriSpeech/MUSDB are not used.

5.6 Polynomial baseline and jump-aware control

RQ: does a cheap classical poly fitter close the S7 residual, and does endpoint pinning show the wrap-jump vs R tradeoff? A seam-local degree-3 polynomial fitter (no learning) reaches canonical $R \approx 0.972$ ($\approx$ no-bake) with wrap-jump essentially unchanged (`poly_baseline.json`). SSM seq models are deferred; the LSTM remains the supervised seq ceiling. An endpoint-pin control (`jump_control.json`) drives wrap-jump to 0 on all / hard subsets while dropping R (all 0.906; hard 0.822). Methods that zero wrap-jump are not the same objective as prolonged R; we do not claim wrap-jump SOTA for the favorite.

5.7 Classical VA seam repairs (BLIT/BLEP, PolyBLEP, BLAMP)

RQ: do the cited VA antialias tools [1–3] beat DualCosine on prolonged R when applied as cycle-local seam residuals?

What each operator changes. Figure 10 shows the family on one high-wrap sine+cliff tile (same seed and tile as Figure 1). **BLIT/BLEP** (Stilson/Smith lineage) treats the wrap as a value step. It subtracts a band-limited step residual (here a raised-cosine BLEP-family lobe scaled by the measured wrap jump) so the tiled click spectrum is softened near the seam.

Table 8: Required seam-local secondary on top-10% wrap-jump tiles (n=410): edge RMSE (lower better). Frozen in `cliff_strata.json`.

Method	top10 edge RMSE
seq LSTM	0.011
N2N sibling-sup.	0.018
neural favorite	0.021
N2N corrupt→corrupt	0.021
seq CNN1D	0.035
no-bake	0.095
`seam_fir3`	0.164
DualCosine	0.990

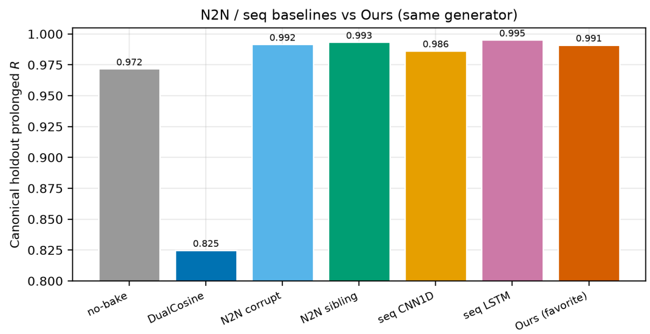

Figure 9: Canonical holdout prolonged R: N2N and seq baselines vs Ours (neural favorite). Train seeds `424242`/`424243` are disjoint from holdout `20260719`.

Table 9: Wavetable-native wrap protocol (Phase F1). Close-seam ideal → open-wrap cliff (seed `20260719`). Mean prolonged R. Primary = ReelSynth export; secondary = AKWF CC0 files.

Method	Export (n=25)	AKWF OA (n=24)
endpoint-pin (mean)	0.939	0.978
`seam_fir3`	0.971	0.975
poly seam (d=3)	0.967	0.970
no-bake	0.967	0.969
neural favorite	0.911	0.970
DualCosine	0.883	0.936
N2N corrupt→corrupt	0.869	0.974
seq CNN1D	0.839	0.962
seq LSTM	0.769	0.948

PolyBLEP (Nam et al.) is a cheap polynomial surrogate of that step residual with fractional-delay support near phase wrap. It edits only a few samples on each side of the discontinuity. **BLAMP** (Esqueda et al.) corrects a slope or corner jump (discontinuity in the first derivative), which matters for triangles, rectification, and hard clipping. On a pure value cliff the slope jump is small, so BLAMP is nearly a no-bake passthrough. **DualCosine** (our classical bake baseline) fades both ends of the stored period with raised-cosine windows. It does not insert a BLEP-style residual. It trades wrap energy for a wide end-fade distortion that prolonged R penalizes on hard cliffs.

Scores. Table 10 reports batch means from `va_seam_blep.json` (seed `20260719`, batch 64). Implementation: `reelsynth/scripts/baselines/va_seam_blep.py` (PolyBLEP matches `osc::va::poly_blep`; BLIT/BLEP uses a raised-cosine step residual; BLAMP applies a poly-BLAMP lobe to the wrap slope jump). PolyBLEP reaches mean $R \approx 0.929$ ($\Delta R \approx +0.104$ vs DualCosine) but trails no-bake / `seam_fir3` / the meta favorite. BLIT/BLEP nearly zeros wrap-jump (0.004) while dropping mean R to 0.868 and collapsing on hard cliffs ($R \approx 0.712$). BLAMP is near-no-bake on this value-cliff geometry (mean $R \approx 0.972$). On the annotated severe tile in Figure 10, tile R is lower still (PolyBLEP ≈ 0.869, BLIT/BLEP ≈ 0.703, DualCosine ≈ 0.603, BLAMP ≈ engine), which matches the hard-cliff strata. Honest reading: these classical tools are scored here, not only cited. They beat DualCosine on mean R in places. None matches the favorite on prolonged R.

5.8 Transfer conditions

RQ: when does hybrid search win vs the classical non-AI board (no-bake / FIR / DualCosine) and supervised nets? Per-cycle ΔR on export, AKWF, Rust `sound_bench`, and sine+cliff holdout (`transfer_failures.json`): favorite win-rate vs DualCosine is high on sine cliffs (1.00) and AKWF (0.88), and moderate on export (0.64) / Rust (0.75); win-rate vs no-bake is high only on sine cliffs (0.92) and collapses on export (0.24) / Rust (0.20). Search value is therefore highest on hard cliffs, on OA tracks where DualCosine collapses, and when compact bake graphs are needed without engine→ideal labels at search time. No-bake/FIR win on mild seams and OOD factory geometry (honest domain shift, reported openly).

5.9 Ablations and compute

RQ: how do search branch-best freezes compare to short isolated GA/PPO budgets, and what compute was used? Table 11 reports two views. Search *branch-best freezes* at the 5k gate come from one hybrid history. Isolated short-budget re-runs (GA-only, PPO-only, GA+PPO, full) use 150 iters on the same search seed with plateau adapt off. Short-budget champions sit below the multi-hour freeze by construction. They are labeled as such. Table 12 summarizes the search-campaign compute budget. Peak VRAM is from an `nvidia-smi` poll during a short full-branch search replay (30 iters) plus a champion forward+FitCell replay. The campaign `history.jsonl` did not log memory.

5.10 Meta-learning approach comparison (5k matched budget)

RQ: under a matched 5,000-evaluation budget with LSTM and xLSTM in the searchable bake vocabulary, how do Random NAS, Cont. CMA-ES, Arch REINFORCE, Aging evolution,

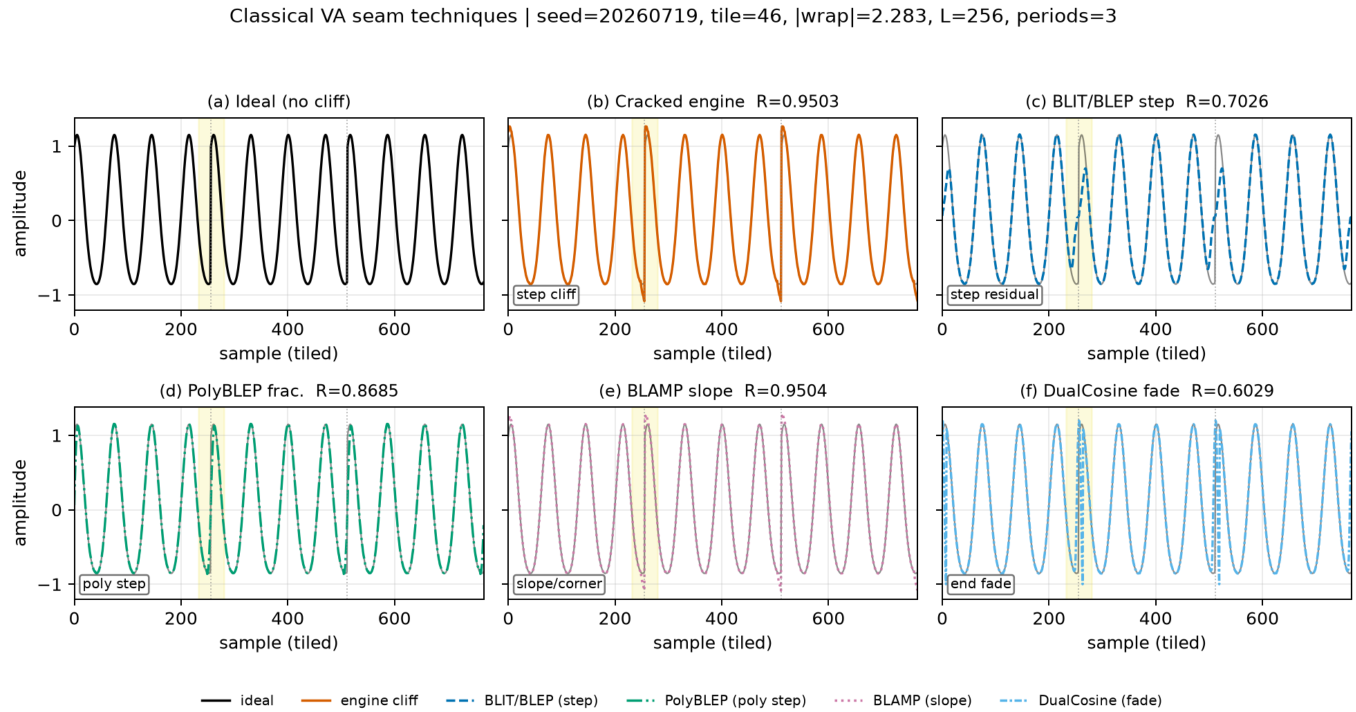

Figure 10: Classical VA seam techniques on one cracked sine+cliff tile (seed 20260719, tile 46, $|wrap|$=2.283, L=256, three tiled periods). Yellow band marks the first seam neighborhood. Panel tile R (1 = best): (a) ideal (unscored sibling). (b) cracked engine R=0.9503. (c) BLIT/BLEP step residual R=0.7026. (d) PolyBLEP fractional-delay residual R=0.8685. (e) BLAMP slope/corner residual R=0.9504 (near no-bake on value cliffs). (f) DualCosine end fade R=0.6029. Colorblind-safe hues with distinct linestyles for B&W print. Batch means appear in Table 10. Accessibility: six-panel comparison of ideal, cracked, BLIT/BLEP, PolyBLEP, BLAMP, and DualCosine.

Table 10: Classical VA seam repairs on the residual protocol (seed 20260719). Canonical holdout: R, SNR/SDR, wrap-jump, edge RMSE. Cliff top-10%: R (n=410). Multi: mean R over 20 generative waveforms.

Method	R	SNR	SDR	jump	edge	top10 R	multi R
BLAMP	0.9718	32.6	32.7	0.873	0.080	0.9667	0.965
no-bake	0.9718	32.6	32.7	0.873	0.080	0.9667	0.965
seam_fir3	0.9610	29.6	29.6	0.438	0.112	0.9429	0.955
PolyBLEP	0.9288	25.9	25.8	0.102	0.205	0.8641	0.924
BLIT/BLEP	0.8680	20.4	20.3	0.004	0.365	0.7116	0.860
DualCosine	0.8249	18.0	17.8	0.292	0.506	0.6574	0.814

Latency (canonical, CUDA): PolyBLEP ≈ 1.43 ms/batch; DualCosine ≈ 1.06; BLIT/BLEP ≈ 123; BLAMP ≈ 54. Frozen in va_seam_blep.json.

and TPE Bayes NAS compare to **Ours (hybrid GA–PPO)**? Table 13 reports champion prolonged R, ΔR vs DualCosine, wall hours, and whether the champion graph used an LSTM or xLSTM block. At the completed gate, **Ours** leads on absolute champion R ($R \approx 0.99146$, $\Delta R \approx +0.17488$ vs DualCosine ≈ 0.81658; ≈ 8.46 h), with Aging evolution second ($R \approx 0.99079$). Neither Ours nor Aging evolution selected an LSTM block in the champion graph. Arch REINFORCE / Aging evolution champions used xLSTM, and TPE used LSTM (vocab presence, not a full-stack claim). Figure 11 is the at-a-glance bar comparison; Figure 12 shows champion-R learning curves (x-axis capped at 5,000). Figure 13 shows a holdout wrap-seam heal (seed 20260719) for the refit Ours champion against no-bake, DualCosine, and ideal sibling $r^{\star}$. Source: meta_approach_compare.json (search seed 1902771841).

5.11 Rust sound_bench transfer (20 tiles)

RQ: does the Python-trained favorite transfer to Rust sound_bench tiles against the classical non-AI board? Table 14 scores no-bake, seam_fir3, DualCosine, other classical rows, and the Python-trained neural favorite on 20 Rust-exported tiles (10 families × 2 seeds). This closes the prior gap where multi-family tables used only Python generative stand-ins. Honest transfer note: the favorite was searched

Table 11: Ablations. Left block: search branch-best freeze (run `gpu-rl-arch-20260719T083019Z`, final iter ≈ 6,922). Right block: isolated short-budget re-runs (150 iters, seed `1902771841`, plateau adapt off). R is prolonged residual on the runner geometry.

Config	Search freeze R	Isolated 150-it R
Full hybrid	0.99093	0.98113
GA-only	0.99016	0.98062
PBT (branch best / n/a isolated)	0.99012	—
PPO-only	0.98924	0.97932
GA+PPO	—	0.98007
combo (branch best)	0.98854	—
NAS (branch best)	0.98677	—
DualCosine (runner)	0.81658	0.81658

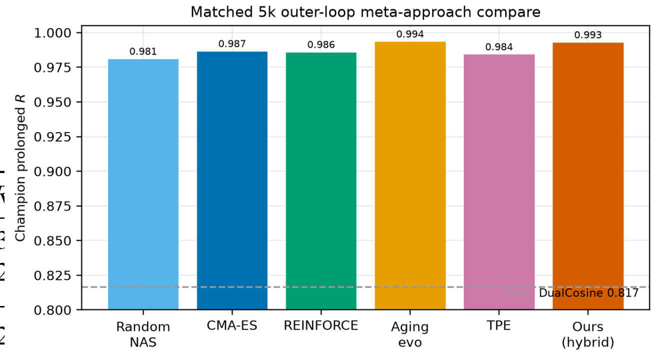

Figure 11: Matched 5k-budget meta-approach comparison (search seed `1902771841`): (a) champion prolonged R; (b) ΔR vs DualCosine (reporting gap only; objective is maximize absolute R toward the ideal sibling). Manuscript names; Okabe–Ito colors with distinct hatches for grayscale print.

Table 12: Compute budget for the reported 5k-gate search freeze (RTX 3090, PyTorch 2.6+cu124, search seed `1902771841`). Arch evaluations ≈ clean iterations.

Item	Value
GPU	NVIDIA GeForce RTX 3090
Elapsed wall time	≈ 7.92 h
Clean iterations / arch evals	≈ 6,922
Population size	12
Final champion R	0.99093
DualCosine baseline (runner)	0.81658
Peak VRAM (nvidia-smi replay)	≈ 3.29 GiB

Peak memory: max `nvidia-smi memory.used` during a 30-iter full-branch search probe (≈ 3.29 GiB) and champion forward+FitCell replay (≈ 3.17 GiB). The multi-hour search process itself left no memory log. Idle GPU occupancy was already ≈ 3.0 GiB from other sessions.

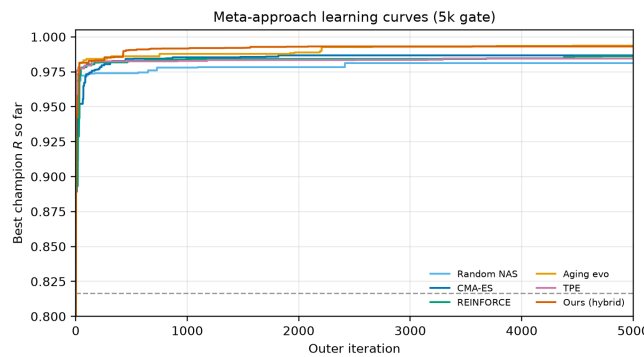

Figure 12: Matched 5k-budget meta-learning comparison: champion residual R versus outer iteration (x-axis capped at 5,000). DualCosine dashed as one classical comparator only. Objective is maximize absolute R toward the ideal sibling; rank vs the classical board in Table 5. Accessibility: Okabe–Ito colors with distinct markers for grayscale print.

Table 13: Matched 5k-budget outer-loop comparison (search seed `1902771841`; LSTM+xLSTM in bake vocab; reward-mode + HP co-tune in Ours). Champion R vs ideal sibling; ΔR vs DualCosine is one classical gap only. Wall-h: per-method CUDA wall time. LSTM?/xLSTM?: whether the champion graph included that block.

Method	Champ R@5k	ΔR vs DualCosine	Wall-h	LSTM?	xLSTM?
Random NAS	0.97816	+0.16157	5.72	no	no
Cont. CMA-ES	0.98468	+0.16810	2.28	no	no
Arch REINFORCE	0.98247	+0.16588	25.93	no	yes
Aging evolution	0.99079	+0.17421	15.02	no	yes
TPE Bayes NAS	0.98106	+0.16448	11.18	yes	no
Ours (hybrid GA–PPO)	0.99146	+0.17488	8.46	no	no

Table 14: Secondary transfer on 20 Rust `sound_bench` tiles (byte-aligned export). Mean±std prolonged R vs the same ideal sibling. Rank by absolute R against classical non-AI rows; ΔR column is vs DualCosine only. Favorite beats DualCosine (mean ΔR+0.063, p≈0.009) but trails no-bake / FIR.

Method	R (20 Rust tiles)	ΔR vs DualCosine
no-bake	0.928 ± 0.084	+0.166
`seam_fir3`	0.896 ± 0.077	+0.134
neural favorite	0.825 ± 0.121	+0.063
classic quadratic	0.788 ± 0.110	+0.026
DualCosine / cosine fade	0.762 ± 0.118	0

on Python `make_batch` sine+cliff geometry. On Rust families it still beats DualCosine (mean ΔR+0.063, Wilcoxon p≈0.009) but trails no-bake and `seam_fir3` on absolute R. The primary SOTA claim remains the Python multi-family matrix.

5.12 Vibrato playback and spectrograms

RQ: do wrap-seam differences remain visible under *dynamic* pitch (slow vibrato), beyond static tiled periods? Figure 23 renders the same holdout cracked period, DualCosine bake, and refit **Ours (hybrid GA–PPO)** champion under shared

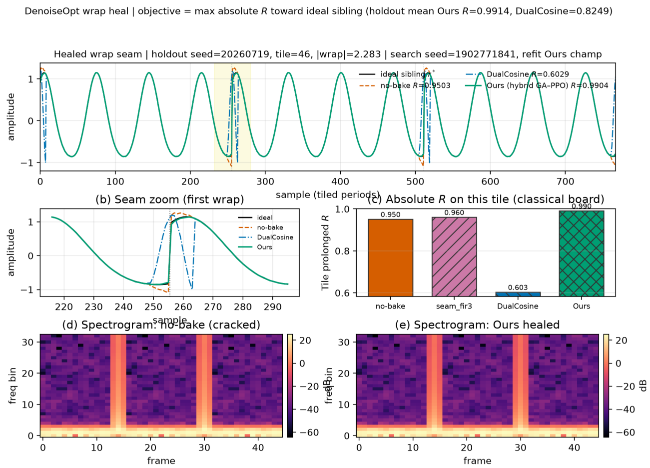

Figure 13: Healed wrap-seam example for the matched-suite **Ours (hybrid GA–PPO)** champion (arch+HP from `hybrid_lstm` summary; FitCell refit with search seed `1902771841`). Holdout tile (seed `20260719`): prolonged waveforms vs ideal sibling, seam zoom, per-tile absolute R vs no-bake / FIR / DualCosine, and spectrograms before/after heal. Cell weights are refit for visualization (suite checkpoints store arch+HP, not fitted `state_dict`); holdout mean R accompanies the tile view. DualCosine ΔR is reporting only.

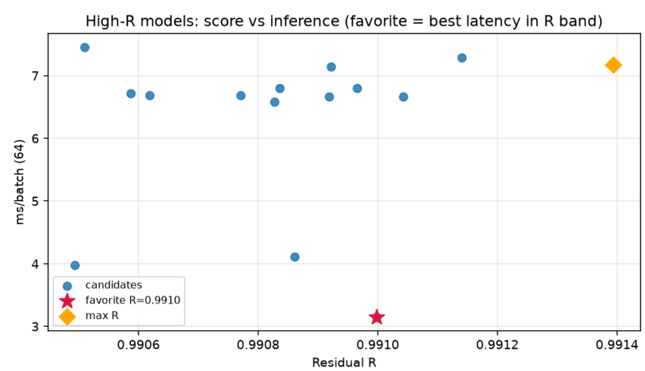

Figure 14: High-R fitted cells on CUDA (batch 64): prolonged residual R versus forward latency (ms/batch). Star marks the favorite (fastest inside the near-max-R band).

sinusoidal vibrato ($\pm 3\%$ depth at 5 Hz around 220 Hz). Panels (a–c) show magnitude spectrograms; (d–e) show DualCosine$-$no-bake and Ours$-$no-bake differences; (g) reports cycle-local prolonged R alongside a modulated-playback residual under the same pitch envelope. On the annotated high-wrap tile, Ours retains high cycle R and high modulated R, while DualCosine remains the weakest classical comparator. This is objective spectrogram / residual evidence only. It is not a formal listening study. Source: `reelsynth/scripts/bench_vibrato_spectrogram.py`; artifacts under `brand/artifacts/vibrato_spectrogram`.

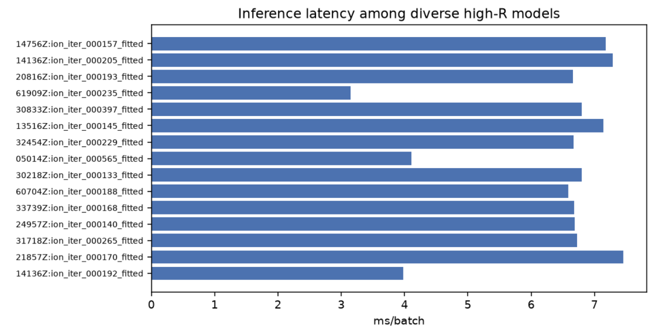

Figure 15: Inference latency (ms/batch, CUDA, batch 64) for diverse high-R fitted models from the inference bench.

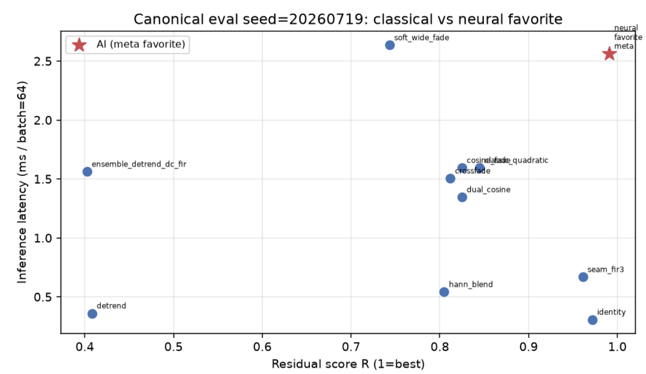

Figure 16: Canonical holdout (seed `20260719`, batch 64): classical bake methods and neural favorite in the R vs CUDA ms/batch plane. Star: favorite.

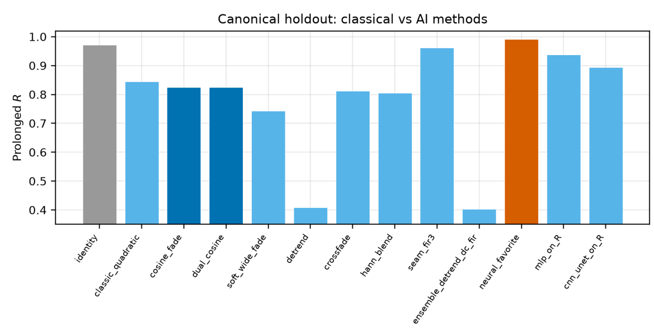

Figure 17: Canonical holdout ranking on prolonged residual R (left) and CUDA latency (right). DualCosine $R \approx 0.825$. Favorite highlighted.

5.13 Downloadable hear samples

As an audible companion to Figure 13, we release five fixed-pitch (A4, 3 s, 44.1 kHz mono) wavetable clips for no-bake, DualCosine, and Ours-healed periods from the matched-suite holdout (`reelsynth/brand/artifacts/meta_approach_compare/h`; rebuild via `scripts/export_meta_hear_samples.py`). Figure 24 shows short waveform excerpts from those WAVs with per-clip absolute R. Readers can download and listen;

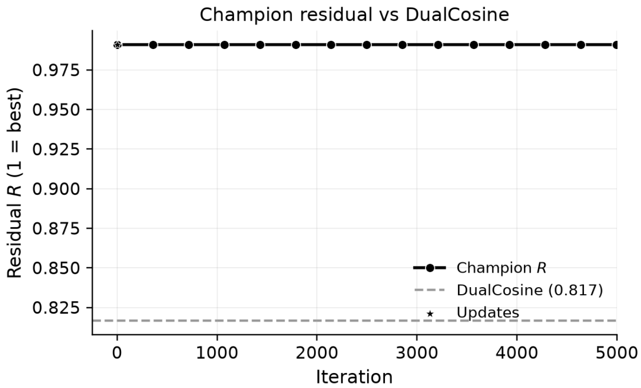

Figure 18: 5k-gate search freeze: champion residual R versus clean search iteration (x-axis capped at 5,000), with DualCosine baseline on the runner geometry (RTX 3090). Accessibility: black line with circle markers; DualCosine as a dashed gray baseline.

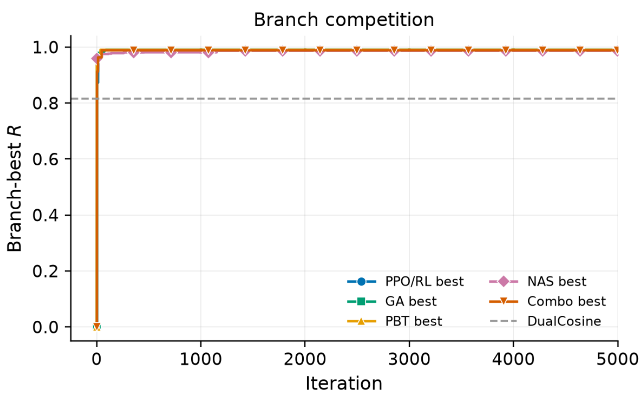

Figure 19: 5k-gate search freeze: per-branch best residual trajectories (PPO/RL, GA, PBT, NAS, combo) versus iteration (x-axis capped at 5,000). Accessibility: colorblind-safe colors plus distinct markers (circle, square, triangle, diamond, inverted triangle) for grayscale print.

we do *not* report formal MOS/MUSHRA or claimed A/B listening scores.

5.14 Compact wavetable diversity gallery

Beyond the synthetic sine+cliff narrative, Figure 25 shows a compact gallery of true ReelSynth-exported factory periods (one mid-morph frame per bank) and Adventure Kid Waveforms (AKWF) CC0 single-cycle files already used by the wrap protocol. Scored means remain in Table 9 / `real_wt_matrix.json`; this panel is diversity evidence only. No new NAS was run for the gallery.

5.15 Sci/eng wrap-heal transfer results

Classical-board disclaimer. Table 15 reports holdout-refit prolonged R on classical / domain-proxy boards only. Deep SOTA rows are listed as *not executed* (Table 16). Synthetic CNC/PMU rows are protocol pilots when real KIT / DataPort downloads are blocked. Artifacts: `reelsynth/brand/artifacts/signal_heal_transfer/`.

Reading. On CWRU, DualCosine and fades hurt vs no-bake; Ours only narrowly beats no-bake (weak transfer on that wrap). On MFPT, Ours leads the classical board ($R{\approx}0.884$) with a small gap over no-bake. On MIT-BIH, DualCosine collapses morphology while Ours leads ($R{\approx}0.879$ vs endpoint-pin $R{\approx}0.780$). On the PTB-XL `records500` (`*_hr`, 500 Hz) subset, Ours leads ($R{\approx}0.640$) but absolute scores are lower than MIT-BIH (different beat construction / domain). Synthetic CNC and PMU proxies show modest Ours gains over DualCosine under the same R protocol; they are not substitutes for logged machine or PMU traces. Primary claim remains cycle-local *wavetable* seam restoration.

Transfer inference latency. Table 17 reports wall-clock inference on transfer holdout batches (device=`cpu`, batch=64, warmup=20, timed=100; includes prolonged residual). N2N uses the wavetable `n2n_corrupt_corrupt` checkpoint zero-shot when present; domain-trained N2N was not run. These timings fill the prior transfer gap (wavetable tables had latency; transfer had R only).

5.16 Listening / spectrogram protocol (main Results)

We place audible and spectrogram evidence in the main Results path (Sections 5.12–5.13), not only an appendix note. Downloadable WAVs (no-bake / DualCosine / Ours, A4, 3 s) and vibrato spectrograms support informal A/B inspection. The planned formal listening protocol (forced-choice wrap-click preference; 44.1 kHz mono; frozen WAV pack) is documented in `reelsynth/brand/artifacts/signal_heal_transfer/LIS` **Formal MOS / MUSHRA scores were not collected in this work.** A future formal listening study would freeze the same WAV pack, recruit listeners under a MUSHRA-style wrap-click impairment scale, and report mean scores with confidence intervals. Until that study runs, perceptual claims remain qualitative and secondary to prolonged R.

Hyperparameter sensitivity (Q3). RQ: how sensitive is hybrid champion R to $\pm 50\%$ shifts of Table 2 knobs? We run a **one-at-a-time sensitivity probe** (not a full 5k re-search). Each config uses 500 outer iterations, seed `1902771841`, and sine+cliff FitCell matched to the meta-compare protocol (batch 48, fit steps 24). Perturbed knobs: population size n, PPO clip ε, fit Adam learning rate, GA second-mutate rate, and PPO entropy coefficient. Table 18 and Figure 28 report champion absolute R versus the Table-2 default (default champ $R \approx 0.9888$; largest $|\Delta R|$ among completed arms ≈ 0.0123 on `lr_m50`). All $11/11$ arms completed under this budget.

This is local sensitivity evidence under the stated budget and seed. The 5k matched-budget ranking (Table 13) remains the primary approach comparison. We do *not* claim that Table-2 defaults are globally optimal. We only report that $\pm 50\%$

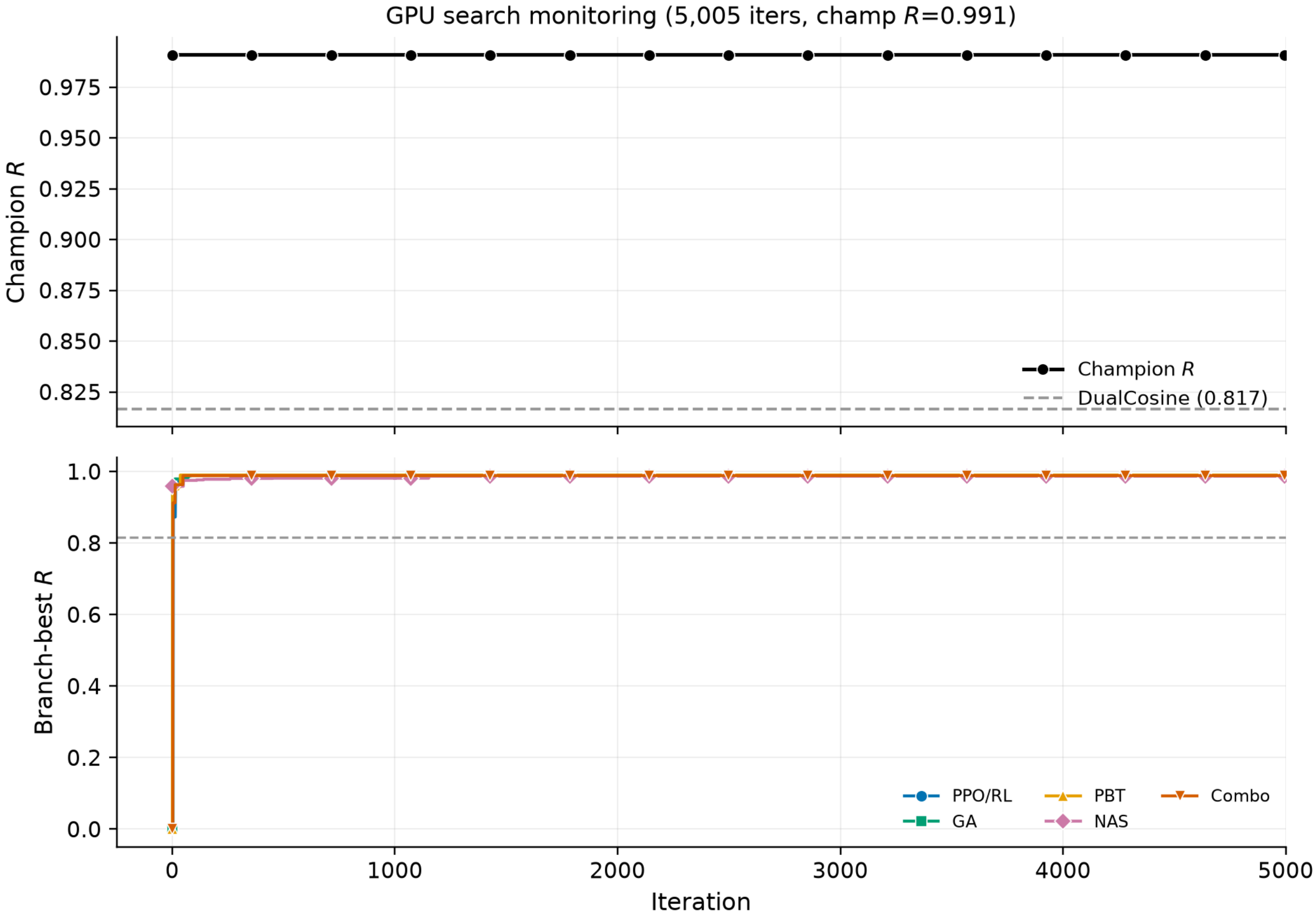

Figure 20: Full-width search monitoring panel at the 5k-gate freeze (first 5,000 iterations): champion R (top) and branch-best trajectories (bottom). Markers and Okabe-Ito colors match Figure 19. Source: `history.jsonl`.

Table 15: Sci/eng wrap-heal transfer: holdout-refit prolonged R (higher better) on classical boards. Ours = `hybrid_lstm`. CWRU/MIT-BIH: 250 outer iters. MFPT/PTB-XL/synth: 150 outer iters (PTB-XL uses PhysioNet `records500 *_hr` @500 Hz subset). Classical board only. No invented deep-SOTA scores.

Method	CWRU	MFPT	MIT-BIH	PTB-XL	synth CNC	synth PMU
Ours (`hybrid_lstm`)	0.8705	0.8842	0.8792	0.6403	0.4675	0.9682
no-bake	0.8675	0.8778	0.6403	0.4503	0.4073	0.9563
endpoint-pin	0.7990	0.7933	0.7796	0.5921	0.4014	0.9605
`seam_fir3`	0.7865	0.8055	0.6615	0.4515	0.4187	0.9477
DualCosine	0.7846	0.7761	0.2009	0.3360	0.4135	0.9393

OAT shifts leave champion R within a small band relative to the DualCosine gap (≈ 0.17 on the 5k gate).

6 Discussion

Where residual mass remains. Across the 100,000-cycle procedural bench and lit-combo family stress tests, wrap error is uneven. The same generator families dominate the residual budget: `nonlinear`, `combo`, `triple_mix`, and `extreme_overlay` show the weakest denoise and residual scores. `harmonic_fft` and `am_fm` are easier. `open_wrap_bias` can look strong on wrap-energy proxies yet still lag on prolonged residual. Cliff reduction and ideal-matched tiling are related objectives with different rankings.

No-bake R and hard-cliff honesty. Canonical no-bake $R \approx 0.97$ looks easy only if R is treated as a perceptual wrap score, or if no-bake is confused with the ideal sibling. No-bake is the unrepaired engine scored against $r^\star$. The ideal withholds the cliff and is never a method output. Under (4), small residual RMS relative to ideal RMS yields high R even when

Figure 21: Per-trial residual scatter by search branch for the first 5,000 iterations at the 5k-gate freeze, with champion overlay. Branch markers match Figure 19.

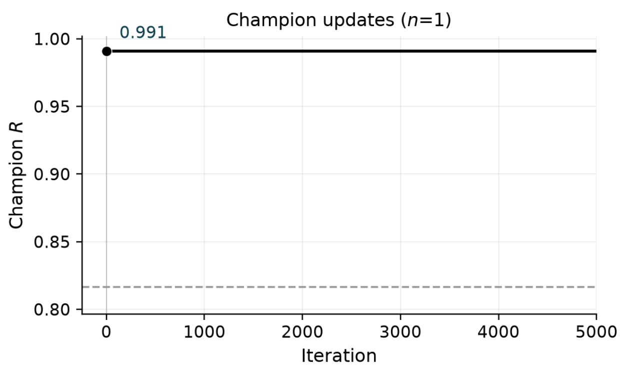

Figure 22: Champion update timeline for the first 5,000 iterations at the 5k-gate freeze. Sparse R labels mark first, midpeak, and latest improvements for twocolumn legibility.

Table 16: Deep / real-data transfer status under this residual protocol. No invented scores.

Item	Status
Cycle-GAN (ECG)	*not executed* (no weights under R)
BeatDiff	*not executed*
Paderborn KAt deep	*not executed* (registration / multi-GB)
Full PTB-XL corpus	deferred (ran `records500 *_hr` subset, n=256 periods)
Real KIT CNC / IEEE PMU	deferred (login walls; synth pilots ran)
Formal MOS/MUSHRA	deferred (Section 5.16)

$|x_0 - x_{L-1}|$ is large. That near-ceiling regime is why tiny ΔR still matters: climbing $0.97 \to 0.991$ uses most of the remaining residual budget, while raw score differences stay small enough that reward shaping and outer-loop hyperparameters must be tuned with the architecture (Section 3.7). Hard-cliff strata make the operative regime explicit. DualCosine drops to $R \approx 0.657$ on the top-10% wrap-jump tiles while the favorite stays near 0.9915. Favorite wrap-jump barely moves toward

Table 17: Transfer-domain inference latency (ms/batch; higher is slower). Batch=64; −− = method unavailable.

Method	CWRU	MFPT	MIT-BIH	PTB-XL	CNC	PMU
no-bake	0.19	0.13	0.22	0.19	0.29	0.20
DualCosine	0.56	0.58	0.70	0.60	0.65	0.63
Ours (`hybrid`)	38.43	40.25	6.62	6.82	26.25	1.64
N2N (WT$\to$dom.)	3.85	3.48	3.86	3.80	3.14	3.16

Table 18: One-at-a-time ±50% hyperparameter sensitivity probe (500 outer iterations, seed `1902771841`, sine+cliff FitCell). Champion absolute R vs Table 2 default. **Sensitivity evidence only** — not a full 5k re-search per HP.

Config	Champ R	ΔR vs default	Iters
`default`	0.98882	+0.00000	500
`pop_size_m50`	0.98136	-0.00746	500
`pop_size_p50`	0.98073	-0.00809	500
`ppo_clip_m50`	0.98159	-0.00723	500
`ppo_clip_p50`	0.99041	+0.00159	500
`lr_m50`	0.97654	-0.01228	500
`lr_p50`	0.98138	-0.00744	500
`ga_mut2_m50`	0.98334	-0.00548	500
`ga_mut2_p50`	0.99002	+0.00120	500
`entropy_m50`	0.98912	+0.00030	500
`entropy_p50`	0.99116	+0.00234	500

zero on that subset. We claim tiled residual repair toward $r^\star$, not endpoint closure.

N2N stress test and transfer conditions. On the sine+cliff holdout, corrupt$\to$corrupt N2N matches the meta favorite within $\approx 10^{-3}$. Sibling-supervised N2N and the LSTM ceiling sit slightly above. That is expected for oracle-label training on the same generator. It does not erase the hybrid search story: DenoiseOpt finds compact bake graphs without engine$\to$ideal supervision at search time, and the hard-cliff DualCosine collapse contrast remains. Win conditions (`transfer_failures.json`): favorite beats DualCosine on sine cliffs and most AKWF / Rust tiles. No-bake and FIR lead on exported factory geometry (domain shift, reported openly). On true ReelSynth exports the searched favorite trails no-bake and FIR. We report that gap. We do not claim universal factory superiority.

Wrap-jump vs edge RMSE. Primary claim remains prolonged R. Wrap-jump is diagnostic only. Edge RMSE is the locked seam-local secondary: favorite reduces it on hard cliffs even when wrap-jump stays flat. Endpoint-pin zeros wrap-jump while hurting R, confirming the objectives differ.

Implication for this paper. Mean R (and mean DualCosine gaps) compress a structured difficulty map. Reporting only the search-campaign mean on synthetic sine cliffs understates how much of the remaining ∼1% sits in a few hard families, and how DualCosine fails when wrap-jump is large. Hard-cliff strata and wavetable-native transfer (true exports + AKWF) are first-class empirical claims here. The contribution targets

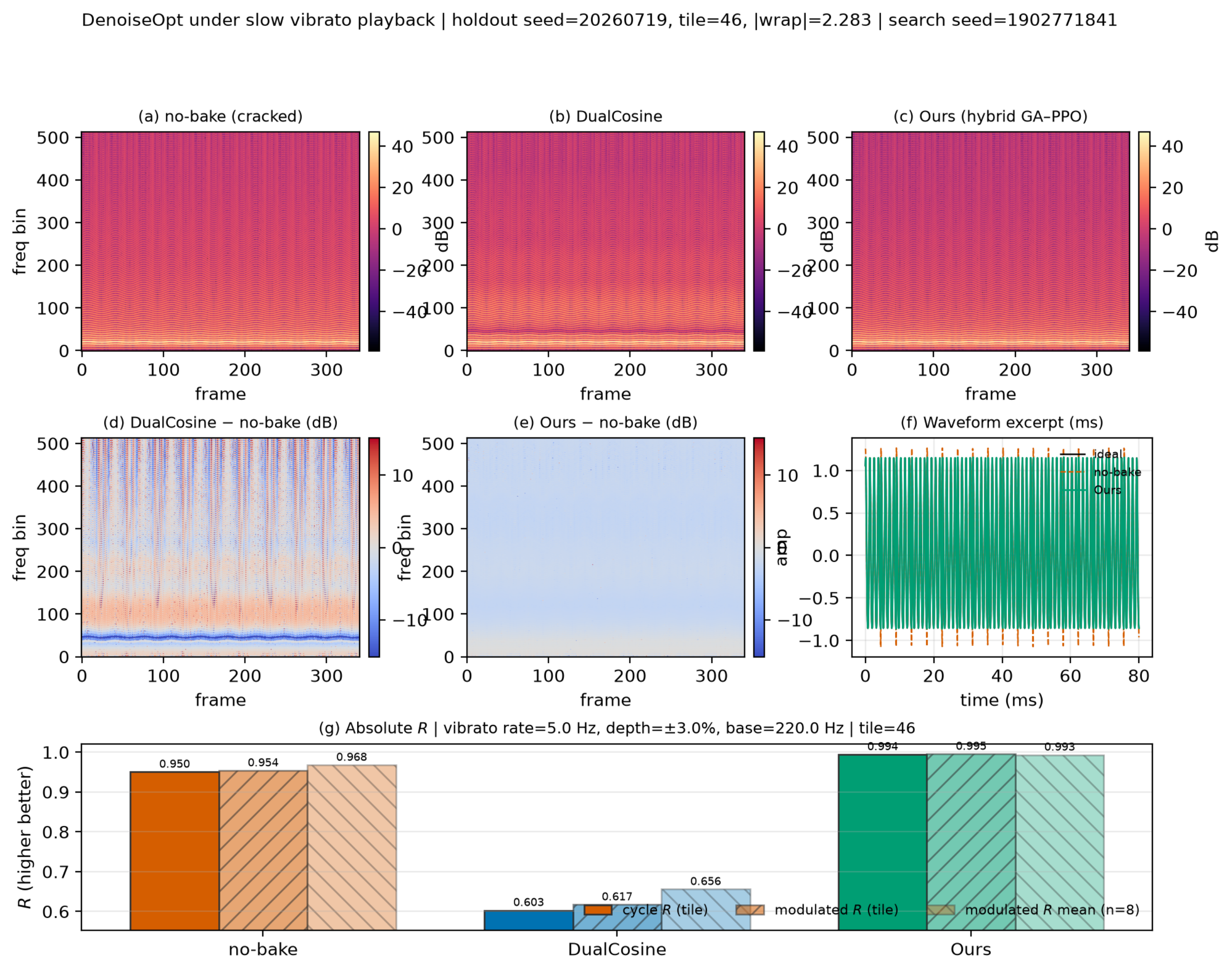

Figure 23: Slow-vibrato spectrogram eval on a high-wrap holdout tile (seed `20260719`). (a–c) Magnitude spectrograms for no-bake, DualCosine, and Ours. (d–e) Spectrogram differences vs no-bake. (f) Short waveform excerpt. (g) Absolute R: cycle-local prolonged residual, modulated-playback residual on the same tile, and mean modulated R over top-wrap tiles. Accessibility: Okabe–Ito hues; difference maps use diverging coolwarm. Formal human A/B scores are out of scope.

DSP / synthesis artifact repair (DAFx / AES / arXiv cs.SD), not general speech-enhancement leaderboards.

Hybrid search vs. plateau. Search traces at the 5k gate show early high champions then long plateaus near $R \approx 0.99$. That plateau is expected under near-ceiling R: further gains are small in absolute units and sensitive to reward shaping, entropy/clip, mutation rates, and plateau-triggered branch reweighting. Plateau adaptation (deeper cells, denser MoE graphs, higher GA/PBT weight) is the operational response inside one campaign. It does not by itself rebalance family-conditional risk. A natural next paper is family-conditional and cliff-conditional meta-learning: train or select denoisers per family, or condition the outer loop on measured wrap statistics, and publish per-family residual curves beside the scalar champion, with ablations of reward shaping under the same compressed-R regime.

Hybrid branches. At the reported 5k-gate search freeze, GA and PBT lead branch-best residuals, with PPO close behind and NAS trailing on this seed (Table 11). These per-branch readings are snapshots from a single hybrid run, not causally isolated single-branch experiments. Isolated short-budget re-runs (150 iterations, same search seed) confirm relative ordering under a fixed compute cap (Table 11). The matched 5k outer-loop comparison (Table 13, Figures 11–12) places Random NAS, Cont. CMA-ES, Arch REINFORCE, Aging evolution, and TPE Bayes NAS beside **Ours (hybrid GA–PPO)** under a shared seed and bake vocabulary that includes searchable LSTM/xLSTM blocks. Ours wins the absolute-R gate ($R \approx 0.99146$) with Aging evolution close be-

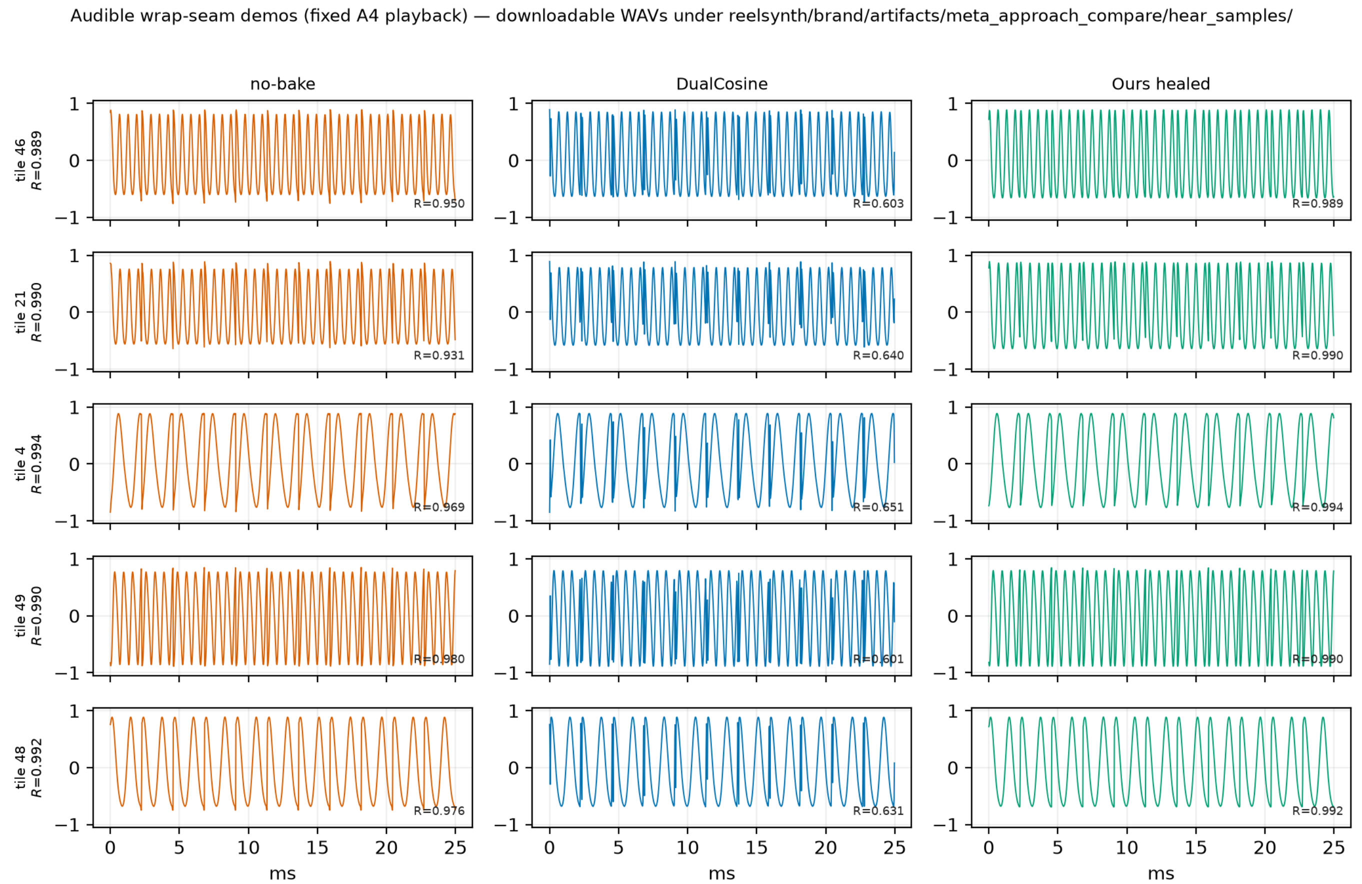

Figure 24: Hear-sample panel from downloadable WAVs under `brand/artifacts/meta_approach_compare/hear_samples/` (five holdout tiles; no-bake / DualCosine / Ours). Waveforms show the first ≈ 25 ms of each clip; R annotations are cycle-local prolonged residuals vs the ideal sibling. Not a formal listening study.

hind ($R \approx 0.99079$). Cont. CMA-ES is the wall-time leader (≈ 2.3 h) at lower champion R. Figure 13 shows the wrap-seam heal qualitatively on holdout geometry. The multi-family SOTA matrix (Table 6) shows the meta favorite ahead of DualCosine on R, SNR, and SDR across twenty Python generative draws, and ahead of no-bake / FIR on absolute R on that same generator. Fixed MLP/CNN-on-R baselines beat DualCosine and trail the searched favorite. On Rust `sound_bench` tiles (Table 14) the same favorite beats DualCosine (mean $\Delta R = +0.063$) and trails no-bake / FIR. That transfer gap matches the narrow Python training geometry.

Review narrative. Peer review asked for Methods clarity, formal ideal-sibling math, broader-than-static eval, and HP justification for meta-search. This manuscript answers with slim IMRaD Methods plus Algorithms 1–9 in the appendix (Section 3, Appendix A); formal Θ and $G(\text{seed}, \text{cliff})$ siblings (Q1–Q2); vibrato spectrograms, downloadable hear WAVs, and a compact real-WT gallery (Sections 5.12–5.14); and a 500-iteration one-at-a-time $\pm 50\%$ HP sensitivity probe (paragraph 5.16) that measures local sensitivity of Table 2 defaults under seed `1902771841`. That probe is not a second 5k re-search and does not re-rank approaches. Sci/eng wrap transfer is reported in main Results as a classical-board pilot (Section 5.15), not a deep-SOTA CWRU/ECG claim. Audible demos support informal inspection only. No formal human listening study was run (protocol in Section 5.16).

7 Tooling and AI use

This section discloses research and writing tools, including AI systems, with scope and intent.

Literature discovery. Open literature was gathered with the Klaut Research Gateway (local OpenAlex / Crossref / arXiv providers) under queries spanning virtual-analog anti-aliasing, unsupervised restoration, HPO/PBT, RL–GA hybrids for NAS, and audio denoise architectures. Harvest JSONs live in the companion meta repository. Source PDFs were downloaded for every bibliography entry (`artifacts/literature_oa/pdfs/`). Paywalled-only items were dropped or replaced.

Paper drafting. Section planning, subsection drafts, and revision passes used Klaut `research_paper_*` tools with local Ollama (`qwen3.5:9b`) when available, under an explicit scientific workflow (plan $\rightarrow$ write $\rightarrow$ revise $\rightarrow$ ex-

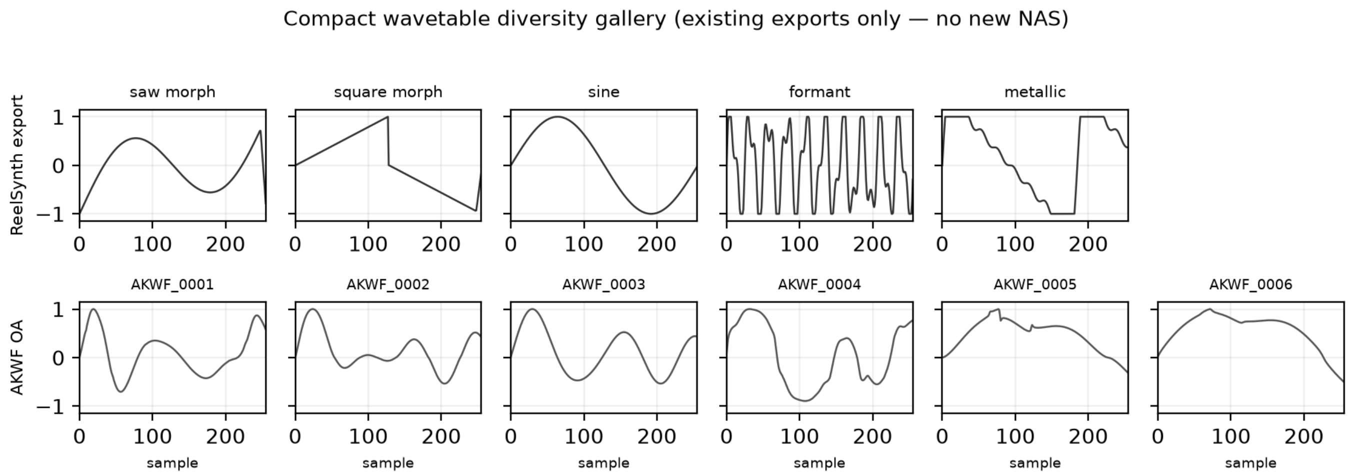

Figure 25: Compact wavetable diversity gallery from existing exports (ReelSynth factory banks, top; AKWF CC0 cycles, bottom). Mid-morph frames per bank; no new meta-search. Quantitative wrap scores: Table 9.

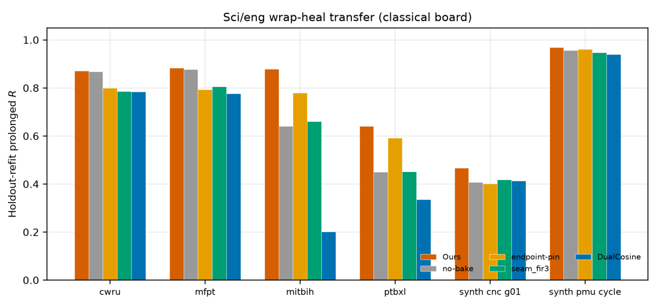

Figure 26: Transfer classical-board summary (holdout-refit R; grouped bars for Table 15). Not a deep-SOTA claim; see Table 16.

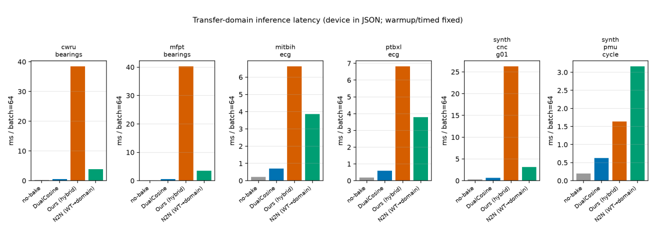

Figure 27: Transfer-domain inference latency (ms/batch). N2N is wavetable-trained zero-shot when available.

port) into an arXiv-style twocolumn template. AI drafts were treated as scaffolding. Claims, numbers, and citations were checked against harvest records, PDF analyses, and measured artifacts. Final prose was edited for DSP terminology and to cut generic filler.

Why AI was used. (1) Scale literature triage across DSP and AutoML without pretending every high-cite hit is in-domain. (2) Accelerate IMRaD scaffolding so human effort focuses on metric honesty (residual vs DualCosine, family hardness, gate vs final 5k-gate search claims). (3) Keep an auditable tool chain (plans, PDF snippet analyses, citation flags) over opaque chat-only writing.

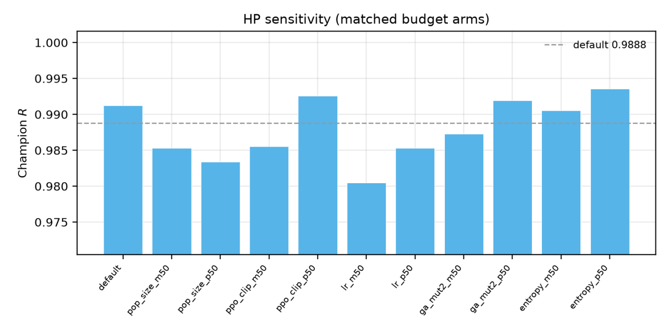

Figure 28: One-at-a-time ±50% hyperparameter sensitivity of hybrid champion R at a 500-iteration budget (seed `1902771841`). Blue bar / dashed line: Table 2 default. This is a **sensitivity probe**, not a full 5k re-search per HP.

Where AI was *not* used. Model fitting, residual scoring, DualCosine baselines, lit-combo / hybrid GPU search, and inference-time benchmarks are conventional code (Rust / PyTorch) with deterministic logs and JSON artifacts. AI did not invent experimental numbers.

Other tools. Rust/`cargo` for the procedural bench and meta binaries. PyTorch CUDA for hybrid GA–PPO search. matplotlib for figures. MiKTeX `pdflatex` for PDF builds. GitHub for versioned papers and artifacts.

Hear samples and vibrato figures. Audible wrap demos: `reelsynth/brand/artifacts/meta_approach_compare/he` (rebuild: `scripts/export_meta_hear_samples.py`). Vibrato spectrogram bench: `scripts/bench_vibrato_spectrogram.py` → `brand/artifacts/vibrato_spectrogram/` (mirrored into `paper/Unsupervised_Wavetable_Seam_Artifact_`

Listening panels for the manuscript are spectrogram + downloadable WAV citations only.

8 Broader impact and reproducibility

Broader impact. This work targets cycle-local wavetable seam repair for virtual-analog / software-instrument workflows. Misuse potential is low. The method does not enhance speech for surveillance, does not remove watermarks from protected media, and operates only on short synthetic periods. Training and search use procedural sine+cliff generators. No private studio stems are required for the reported tables. The main resource cost is GPU time on a consumer RTX 3090 (order of hours for the 5k-gate freeze, Table 12). Authors should not over-claim speech enhancement or production mastering quality from these metrics.

Reproducibility. Holdout seed `20260719`. Hybrid search seed `1902771841`. Primary code: `https://github.com/reeldemo/reelsynth` (`scripts/overnight_gpu_rl_arch.py`, `scripts/bench_sota_matrix.py`, `scripts/metrics_snr_sdr.py`, `scripts/bench_va_seam_blep.py`). Paper companion: `https://github.com/reeldemo/denoise-opt-meta`. Dependencies for the CUDA benches: Python 3.12, PyTorch 2.6+CUDA 12.4, NumPy/matplotlib. Rust/`cargo` builds the separate `sound_bench` hardness probe. JSON artifacts under `brand/artifacts/` (`sota_matrix.json`, `canonical_eval_dataset/`, search `history.jsonl`, `va_seam_blep.json`) freeze the numbers in Section 5. PESQ/STOI are omitted for non-speech cycles (domain mismatch). MUSHRA was not run.

8.1 Funding

This research received no external funding.

8.2 Author contributions (CRediT)

Julian M. Kleber: Conceptualization, Methodology, Software, Investigation, Formal analysis, Data curation, Writing (original draft), Writing (review and editing), Visualization.

8.3 Data availability

Frozen evaluation artifacts (JSON matrices, search history, fitted champions) are published with the companion repositories: `https://github.com/reeldemo/reelsynth` under `brand/artifacts/`, and `https://github.com/reeldemo/denoise-opt-meta` under `artifacts/` and this paper folder's `figures/`. All reported tables regenerate from those files plus the scripts named above. No private studio recordings were used.

8.4 Competing interests

None declared.

9 Limitations

Scope and evaluation limits. The prolonged residual R is defined against a procedural no-cliff sibling and does not capture perceptual quality directly. Formal listening tests (MOS, MUSHRA) and claimed A/B listening scores are outside the scope of this work. We ship spectrograms and downloadable WAVs for informal audition only. PESQ and STOI are excluded because the primary cycles are non-speech synthetic wavetable tiles. Speech-quality metrics on sine+cliff families would be a domain mismatch. MUSHRA was not run (requires human listeners). Any perceptual or speech-proxy secondary study would need separately labeled open-access speech snippets and remains out of scope for this narrow-seam paper.

- Residual uses a procedural ideal (no-cliff sibling). Perceptual MOS and formal listening panels are out of scope.
- Bake metrics leave formal listening tests and speech quality metrics to separate studies.
- **PESQ / STOI / MUSHRA deferred.** Primary cycles are non-speech wavetable tiles. Speech metrics would domain-mismatch sine+cliff families, so PESQ/STOI numbers are absent here (none invented on synthetic tiles). MUSHRA needs human listeners and was *not run*.
- **No LibriSpeech / MUSDB.** Wavetable-native realism uses true ReelSynth-exported factory periods (primary) and external AKWF CC0 single-cycle WAVs (secondary) under the wrap protocol. Procedural stand-ins are tertiary smoke only. Speech/music enhancement corpora remain off by design.
- **HP sensitivity is not a 5k re-search.** Paragraph 5.16 reports one-at-a-time $\pm 50\%$ probes at 500 outer iterations (seed `1902771841`). It measures local sensitivity of Table 2 defaults. It does not re-prove the matched 5k approach ranking.
- **Classical-board transfer only where deep models did not run.** Section 5.15 elevates CWRU/MFPT/MIT-BIH/PTB-XL-subset and synthetic CNC/PMU pilots into main Results. Cycle-GAN, BeatDiff, Paderborn deep, real KIT/PMU streams, and formal MOS remain not executed (Table 16).
- Classical poly seam fitter is reported; lightweight SSM seq models are deferred (timebox); LSTM remains the supervised seq ceiling.
- On exported factory / Rust tiles the meta favorite can trail no-bake or FIR (domain shift). Transfer win-rates are frozen in `transfer_failures.json`; we do not claim universal factory superiority.

- The CUDA search runner scores Python `make_batch` sine+cliff draws. A separate Rust `sound_bench` export of 20 family tiles (byte-aligned generator) is a secondary transfer block. The Python generative SOTA matrix is the primary multi-family table.
- Per-trial architecture width is capped for search throughput. Full Demucs-/diffusion-scale stacks were screened out (domain mismatch vs L=256 periodic seams).
- We do not claim wrap-closure or hybrid-search convergence theorems. Empirical gains are protocol-bound. Endpoint-pin shows wrap-jump can be zeroed while R falls.
- The 5k-gate freeze reports measured champion and branch-best residuals. Isolated GA/PPO/GA+PPO/full re-runs use a short labeled budget (same search seed). Those short runs are a different protocol from the multi-hour freeze.
- Classical VA/BLEP lineage cites are OA author or proceedings PDFs (`stilson1996`, `nam2009polyblep`, `esqueda2016blamp`). Cycle-local BLIT/BLEP, PolyBLEP, and BLAMP seam repairs are now scored (`va_seam_blep.json`); they beat DualCosine on prolonged R in places but trail no-bake / FIR / the meta favorite on the value-cliff geometry.

9.1 Outlook: follow-up publication

A self-contained follow-up can treat *which sounds fail* as the primary object of study: stratified residual heatmaps over the ten procedural families, cliff-size conditioning, and family-specialist or mixture-of-experts denoisers selected by the same RL–GA outer loop. That paper would cite the present mean-R hybrid protocol as the baseline and ask whether beating $R > 0.999$ is mainly an architecture problem or a coverage problem over hard families.

10 Conclusion

DenoiseOpt casts wavetable wrap discontinuity as cycle-local seam repair under a prolonged residual R, with a hybrid GA–PPO(+PBT) outer loop over bake cells. At the 5k-gate search freeze the champion reaches $R \approx 0.9909$ versus DualCosine ≈ 0.8166. On the frozen holdout a compact favorite reaches $R \approx 0.991$ versus DualCosine $R \approx 0.825$. Across twenty multi-family draws, mean $\Delta R \approx +0.162$ (Wilcoxon $p \approx 8.9\times10^{-5}$). Hard-cliff strata keep favorite $R \approx 0.9915$ while DualCosine falls to ≈ 0.657. Same-generator N2N / seq baselines bound the supervised ceiling. Wavetable-native transfer covers factory and OA instrument cycles under the wrap protocol. A classical-board CWRU / MFPT / MIT-BIH / PTB-XL-subset wrap transfer, plus synthetic CNC/PMU pilots, appears in Section 5.15. Deep SOTA weights and formal MOS were not executed. Scope remains wavetable seam restoration for DSP / synthesis venues. Code: `https://github.com/reeldemo/reelsynth`, `https://github.com/reeldemo/denoise-opt-meta`.

Algorithm 1 ResidualScore

Require: Ideal cycle $r^\star$, baked cycle y, periods N
1: $I \leftarrow \mathrm{Tile}(r^\star, N)$
2: $Y \leftarrow \mathrm{Tile}(y, N)$
3: $r_{\mathrm{rms}} \leftarrow \mathrm{RMS}(Y - I)$
4: $i_{\mathrm{rms}} \leftarrow \max(\mathrm{RMS}(I), \varepsilon)$
5: **return** $\mathrm{clamp}(1 - r_{\mathrm{rms}}/i_{\mathrm{rms}}, 0, 1)$

Algorithm 2 DualCosine baseline

Require: Engine cycle x
1: Fade both ends of x with raised-cosine windows of width W
2: **return** $\mathrm{ResidualScore}(r^\star, x_{\mathrm{faded}}, N)$

Algorithm 3 No-bake (passthrough) control

Require: Engine cycle x (open-wrap cliff present)
1: **return** $\mathrm{ResidualScore}(r^\star, x, N)$ $\triangleright$ no operator: $x \mapsto x$

Acknowledgments

Compute for the CUDA hybrid search campaign used a local NVIDIA GeForce RTX 3090 workstation. Open literature was retrieved via the Klaut Research Gateway under open-access PDF constraints. No external collaborators contributed text, code, or experimental numbers for this preprint.

A Algorithms

Companion pseudocode for the residual score, classical controls, FitCell, and hybrid outer-loop branches. Canonical narrative lives in Section 3; these listings mirror `docs/PSEUDOCODE.md` and the CUDA search runner (`overnight_gpu_rl_arch.py` / `denoise_meta_evo.py`).

B Transfer artifact pointer

Full sci/eng wrap-heal transfer protocol, classical-board tables, and deep-SOTA status rows are in Sections 4.2–5.15 of the main text. Algorithms for the residual score and hybrid loop remain in Appendix A. Raw JSON and figures: `reelsynth/brand/artifacts/signal_heal_transfer/`.

Algorithm 4 MoE soft gating over bake experts

Require: Cycle x, expert cells $\{E_k\}_{k=1}^{K}$, gate logits $g \in \mathbb{R}^K$

$w \leftarrow \mathrm{softmax}(g)$
$y \leftarrow \sum_{k=1}^{K} w_k E_k(x)$
return y

Algorithm 5 FitCell (inner loop)

Require: Architecture cfg, fit steps T, batch size B

cell $\leftarrow$ SeamCell(cfg); opt $\leftarrow$ Adam(cell.params)
prev $\leftarrow$ null; patience $\leftarrow 0$
for $t = 1, \ldots, T$ **do**
  ideal, eng $\leftarrow$ MakeBatch(B)
  out $\leftarrow$ ApplyOps(eng, cell, cfg.ops)
  $R \leftarrow$ mean ResidualScore(ideal, out)
  loss $\leftarrow 1 - R$
  if finite(loss) **then** AdamStep(opt, loss)
  end if
  if prev $\neq$ null and $|\text{prev} - R| / \max(|\text{prev}|, \varepsilon) < 10^{-4}$ **then**
    patience $\leftarrow$ patience $+1$; **if** patience ≥ 3 **then break**
  else
    patience $\leftarrow 0$
  end if
  prev $\leftarrow R$
end for
return cell, R

Algorithm 6 GA tournament step

Require: Population P, tournament size k, crossover rate p_c, mutation rate p_m

Sample k genomes; parents $\leftarrow$ top-2 by R
child $\leftarrow$ Crossover(parents) with prob. p_c, else clone elite
Mutate(child) with rate p_m (ops, depth, cell type, θ)
Fit and evaluate child; insert into P (replace worst if full)
return updated P

Algorithm 7 PPO architecture update

Require: Actor–critic π_ϕ, clip ε, experience buffer of mutate actions

Sample action a editing ops / hyperparameters from π_ϕ
Fit proposed cfg; observe centered advantage $\rho = R - R_{\mathrm{DualCosine}}$ ▷ objective remains maximize R
Advantage $\hat{A}$ from critic; ratio $r_\phi = \pi_\phi(a)/\pi_{\phi_{\mathrm{old}}}(a)$
Update ϕ with clipped surrogate $\min(r_\phi \hat{A}, \mathrm{clip}(r_\phi, 1-\varepsilon, 1+\varepsilon)\hat{A})$
return updated π_ϕ

Algorithm 8 PBT exploit–mutate

Require: Population P, exploit quantile q, mutate scale σ

Rank P by R; for each bottom member, copy hyperparameters from a top-q elite
Perturb copied hyperparameters by multiplicative noise of scale σ
Refit briefly; write back into P
return updated P

Algorithm 9 HybridMetaSearch

Require: Iteration budget T_{out}, population size n

$P \leftarrow$ InitPopulation(n); $\pi \leftarrow$ ActorCritic(); champ $\leftarrow$ null; boredom $\leftarrow 0$
for $it = 1, \ldots, T_{\mathrm{out}}$ **do**
  branch $\leftarrow$ Rotate(ppo, ga, pbt, nas, combo)
  cfg, hp $\leftarrow$ Propose(branch, P, π)
  cell, $R_{\mathrm{fit}} \leftarrow$ FitCell(cfg, hp.fit_steps)
  $R_{\mathrm{eval}} \leftarrow$ EvalCell(cell)
  $R \leftarrow \frac{1}{2} R_{\mathrm{fit}} + \frac{1}{2} R_{\mathrm{eval}}$ (+ optional depth/MoE bonus)
  UpdatePopulation(P, cfg, R); PPOUpdate(π, centered $\rho = R - R_{\mathrm{DualCosine}}$)
  if $R >$ champ.R **then** champ $\leftarrow$ (cfg, cell, R); boredom $\leftarrow 0$
  else boredom $\leftarrow$ boredom $+1$
  end if
  if boredom ≥ 1000 **then** PlateauAdapt; boredom $\leftarrow 0$
  end if
end for
return champ

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